FOOD COMPOSITION AND ADDITIVES

Determination of Insoluble, Soluble, and Total Dietary Fiber (CODEX Definition) by Enzymatic-Gravimetric Method and Liquid Chromatography: Collaborative Study

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A method for the determination of insoluble (IDF), soluble (SDF), and total dietary fiber (TDF), as defined by the CODEX Alimentarius, was validated in foods. Based upon the principles of AOAC *Official Methods*SM 985.29, 991.43, 2001.03, and 2002.02, the method quantitates water-insoluble and water-soluble dietary fiber. This method extends the capabilities of the previously adopted AOAC *Official Method* 2009.01, Total Dietary Fiber in Foods, Enzymatic–Gravimetric–Liquid Chromatographic Method, applicable to plant material, foods, and food ingredients consistent with CODEX Definition 2009, including naturally occurring, isolated, modified, and synthetic polymers meeting that definition. The method was evaluated through an AOAC/AACC collaborative study. Twenty-two laboratories participated, with 19 laboratories returning valid assay data for 16 test portions (eight blind duplicates) consisting of samples with a range of traditional dietary fiber, resistant starch, and nondigestible oligosaccharides. The dietary fiber content of the eight test pairs ranged from 10.45 to 29.90%. Digestion of samples under the conditions of AOAC 2002.02 followed by the isolation, fractionation, and gravimetric procedures of AOAC 985.29 (and its extensions 991.42 and 993.19) and 991.43 results in quantitation of IDF and soluble dietary fiber that precipitates (SDFP). The filtrate from the quantitation of water–alcohol-insoluble dietary fiber is concentrated, deionized, concentrated again, and analyzed by LC to determine the SDF that remains soluble (SDFS), i.e., all dietary fiber polymers of degree of polymerization = 3 and higher, consisting primarily, but not exclusively, of oligosaccharides. SDF is calculated as the sum of SDFP and SDFS. TDF is calculated as the sum of IDF and SDF. The within-laboratory variability, repeatability SD (s_r), for IDF ranged from 0.13 to 0.71, and the between-laboratory variability, reproducibility SD (s_R), for IDF ranged from 0.42 to 2.24. The within-laboratory variability s_r for SDF ranged from 0.28 to 1.03, and the between-laboratory variability s_R for SDF ranged from 0.85 to 1.66. The within-laboratory variability s_r for TDF ranged from 0.47 to 1.41, and the between-laboratory variability s_R for TDF ranged from 0.95 to 3.14. This is comparable to other official and approved dietary fiber methods, and the method is recommended for adoption as Official First Action.

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Dramatic increases in the use of fiber analyses in fiber research and the marketing, research, and development of food products have accompanied increases in public

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awareness of the health benefits of high fiber foods over the past several decades. Relationships of particular health benefits to the insoluble or soluble fractions of dietary fiber have been established (1). As researchers have discovered and elucidated additional dietary fiber sources, there has been a need to update the definition of dietary fiber and update the methodologies that support the definition. AOAC INTERNATIONAL has been a leader in providing *Official Methods of Analysis*SM consistent with the state of dietary fiber science (2). In the 1970s, Trowell and fellow dietary fiber researchers (3–6) published a definition later adopted as consensus by AOAC INTERNATIONAL following an international survey by Prosky et al. (7) in the late 1970s.

The Trowell et al. (6) definition.—Dietary fiber consists of the plant polysaccharides and lignin which are resistant to hydrolysis by digestive enzymes of man. This definition defines a macro constituent of foods which includes cellulose, hemicellulose, lignin, gums, modified celluloses, mucilages, oligosaccharides, and pectins, along with associated minor substances, such as waxes, cutin, and suberin.

As the science of dietary fiber has advanced, AOAC *Official Methods* **985.29** and its extensions **991.42**, **993.19** (8), and methods that give equivalent results (**991.43**, **992.16**, **993.21**, and **994.13**; 9) adequately quantitated the known dietary fiber food fractions at the time of the Trowell et al. definition. As the science of dietary fiber has advanced further, these methods became insufficient to quantitate all the dietary fiber components now known, but which were not necessarily known or characterized at the time of their adoption. Advances in understanding the complexity of dietary fiber, and that food components, such as resistant starch, fructans, polydextrose, and resistant maltodextrins, are part of dietary fiber in the diet has led to an updated definition adopted by the CODEX Alimentarius Commission (CAC) and an updated total dietary fiber (TDF) method (AOAC **2009.01**). Recently, international authorities on dietary fiber definition, working through the CODEX Committee on Nutrition and Foods for Special Dietary Uses (CCNFSDU), have updated the terminology of the dietary fiber definition (10) and recommended adoption by the CAC. The CAC adopted this definition during its 2009 session. The wording underwent minor editing of the first footnote during the 2009 session of the CCNFSDU and the 2010 session of the CAC.

CODEX defines dietary fiber as carbohydrate polymers (*see a* below) with 10 or more monomeric units (*see b* below), which are not hydrolyzed by the endogenous enzymes in the small intestine of humans and belong to the following categories: edible carbohydrate polymers naturally occurring in the food as consumed; carbohydrate polymers, which have been obtained from food raw material by physical, enzymatic, or chemical means and which have been shown to have a physiological effect of benefit to health as demonstrated by generally accepted scientific evidence to competent authorities; and synthetic carbohydrate polymers which have been shown to have a physiological effect of benefit to health as demonstrated by generally accepted scientific evidence to competent authorities.

(*a*) Dietary fiber derived from a plant origin may include fractions of lignin and/or other compounds associated with polysaccharides in the plant cell walls. These compounds also may be measured by certain analytical method(s) for dietary fiber. However, such compounds are not included in the definition of dietary fiber if extracted and reintroduced into a food.

(*b*) Decision on whether to include carbohydrates of three to nine monomeric units should be left up to national authorities.

In 2007, McCleary (11) described a method of extended enzymatic digestion at 37°C to simulate human intestinal digestion followed by gravimetric isolation and quantitation of high molecular weight dietary fiber (HMWDF) and the use of LC to quantitate low molecular weight soluble dietary fiber (LMWSDF). The use of the terms HMWDF and LMWSDF in this context are a bit of a misnomer, since the method does not quantitate dietary fiber on the basis of MW, but rather on the basis of solubility in a solution of one part water and four parts alcohol. Thus, HMWDF actually consists of dietary fiber that is insoluble in or precipitates in the water–alcohol mixture, and LMWSDF is dietary fiber that remains soluble. The method reported here quantitates the water-insoluble dietary fiber (IDF) separately from water-soluble dietary fiber (SDF), including the dietary fiber that precipitates (SDFP) in the presence of 78% aqueous ethanol or industrial methylated spirits (IMS), and the dietary fiber that remains soluble (SDFS) in 78% aqueous ethanol (formerly termed LMWSDF). The method thus quantitates the complete range of water-insoluble/soluble fractions (by inclusion of the fractionation steps of AOAC **991.42** and **993.19**) of dietary fiber components from resistant starch, by utilizing the digestion conditions of AOAC **2002.02**, to digestion-resistant oligosaccharides, by incorporating the deionization and LC procedures similar to those of AOAC **2001.03**. A further option for increased laboratory productivity by using online, simultaneous deionization and LC quantitation has recently been published (12) and incorporated as an option in the method.

With the possible exception of synthetic carbohydrate polymers, the food components included in the CODEX definition match those of the more broadly stated Trowell et al. (6) definition. Therefore, the currently adopted *Official Methods* can be readily applied, along with the same issues of extra expense for multiple assays, the need for mathematical summations, and special procedures needed to avoid double counting. The single *Official Method* that quantitates the entirety of components included in the Trowell and the CODEX definitions was validated and adopted as AOAC **2009.01**. Extending that method to quantitate the IDF and SDF fractions is the important next step to provide a continuum of methodology to meet the needs of the dietary fiber research, regulation, and labeling communities. The principles for delineating the IDF and SDF fractions of dietary fiber were first validated for AOAC *Official Methods* **991.42** and **993.19** and later incorporated as part of the validation of AOAC **991.43**. Applying these principles to AOAC **2009.01** provides a robust, time-proven method suitable for the purpose.

Collaborative Study

Precollaborative Ruggedness Testing

Based on the successful study that resulted in the adoption of *Official Method* **2009.01**, it was determined that additional precollaborative ruggedness testing of this method was not needed.

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Table 1. Collaborative study data for insoluble dietary fiber (%IDF reported by laboratories)[a]

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1	25.90	25.10	5.30	5.00	10.10	10.80	1.20	1.20	8.50	8.20	13.90	14.20	11.80	11.70	10.30	10.30
2	24.59	26.00	4.91	4.33	7.64	7.00	1.26	1.33	7.81	7.88	12.06	13.32	9.48	10.25	9.40	9.34
3	26.60	25.01	4.72	4.51	7.30	9.03	1.21	1.11	7.79	7.65	11.31	10.45	10.90	10.76	8.86	8.57
4	25.39	25.18	5.09	4.91	12.40	13.15	1.16	1.19	8.31	8.32	10.59	10.81	13.33	13.46	10.24	10.27
5	27.28	27.28	5.43	5.76	10.38	11.16	1.68	1.89	9.20	9.33	12.60	13.02	13.25	13.98	11.64	11.10
6	26.91c	30.52c	4.13	4.29	10.40	8.73	0.54	0.52	7.68c	11.08c	37.94sg	38.32sg	16.10dg	16.93dg	15.77sg	15.04sg
7	24.70	25.51	5.35	5.16	8.78	8.71	1.23	1.51	8.44	8.65	11.55	11.46	11.92	11.99	10.31	10.82
8	26.15	25.81	5.02	5.42	9.30	11.35	0.91	0.94	12.38c	10.34c	11.69	13.16	25.81c	12.24c	13.65c	10.14c
9	23.95	24.44	4.43	4.25	7.27	7.50	0.71	0.28	8.18	7.66	10.29	10.38	9.95	10.10	8.90	8.77
10	27.77	27.58	4.06	3.89	10.39	10.21	0.07	0.20	8.08	7.85	11.40	10.89	16.25dg	16.64dg	10.09	10.10
11	27.32	27.09	5.16	5.15	12.41	11.61	1.08	1.18	8.74	8.73	11.03	11.50	11.79	11.53	10.60	10.27
12	24.41	24.54	4.48	4.40	3.42	4.87	1.00	0.96	7.56	8.02	10.76	10.43	11.65	10.62	10.10	9.12
13	23.63	24.00	4.81	4.74	8.18	8.42	1.01	1.06	7.99	8.18	11.49	11.82	12.23	11.92	9.36	9.77
14	26.24	26.97	5.08	4.91	10.69	10.78	1.18	1.54	8.58	8.97	11.52	10.87	11.40	11.85	10.06	10.40
15	25.7	26.5	4.6	4.4	7.6	7.3	1.2	1.1	8.1	8.0	*[b]	*	10.3	10.3	9.6	9.1

[a] A = Cabbage, B = mixed grains with apple flakes, C = chocolate with fructooligosaccharides, D = biscuits containing fructooligosaccharides, E = defatted cookies with oat graham, polydextrose, RS2 starch, F = peanuts, G = oat bran, H = whole wheat bread with 2% α-cyclodextrin; c = Cochran test outlier on insoluble dietary fiber; sg = Single Grubbs test outlier on insoluble dietary fiber; dg = Double Grubbs test outlier on insoluble dietary fiber.

[b] * = Laboratory reported no result for this sample.

Table 2. Collaborative study data for soluble dietary fiber (%SDF reported by laboratories using manual deionization)[a]

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1	3.00	2.72	4.57	4.80	16.95	17.52	10.58	10.82	12.53	12.02	3.22	3.15	11.10	11.34	6.95	7.51
2	3.34	3.32	5.70	5.71	16.37	16.56	11.00	10.88	11.81	11.65	3.70	3.53	10.69	10.49	8.24	8.63
3	4.51c	8.22c	7.04	6.72	18.01	20.90	11.10	12.18	15.51	16.98	6.58	5.15	12.44	14.83	11.09	11.36
4	4.98	5.86	6.75	7.27	17.84	18.30	11.17	11.01	13.61	14.25	4.01	6.88	12.14	12.76	10.24	9.24
5	3.05	4.31	5.76	5.80	17.17	17.08	10.12	11.66	12.66	12.67	10.26c	4.58c	11.91	11.57	8.13	7.82
6	5.76	5.14	5.78c	7.41c	18.79	18.77	12.84	12.97	13.55	14.04	5.10	4.74	13.32	12.70	11.31c	14.74c
7	4.26	3.39	5.56	5.52	15.42	15.81	9.11	9.04	10.63	10.54	2.87	3.34	11.14	11.74	7.92	7.58
8	2.72	2.94	4.37	4.53	16.67	18.16	9.15	9.29	11.77c	14.60c	7.94	8.13	8.67	9.25	9.14	10.33
9	4.33	4.31	5.22	4.94	16.87	17.21	10.17	9.92	13.12	13.07	3.77	3.91	11.14	11.04	8.51	7.87
10	3.87	4.57	5.82	6.88	18.60	14.8	11.42	11.25	14.41	13.85	5.16	7.62	12.89	11.79	7.19	7.02
11	3.98	4.33	6.16	6.58	17.89	17.95	10.80	11.52	13.53	13.61	4.24	4.43	11.63	12.69	8.68	8.17
12	2.66	1.91	5.19	5.70	18.55	16.12	11.02	12.35	11.68	10.29	3.20	2.99	8.87	6.76	6.12	7.31
13	4.56	4.04	5.26	5.25	18.42	18.64	10.99	11.35	13.22	14.49	4.56	4.93	12.17	12.23	8.01	8.66
14	3.71	3.69	5.11	4.69	16.19	16.12	10.51	9.99	11.67	11.69	2.31	3.17	10.16	11.14	7.59	6.52
15	2.44	3.12	4.63	4.64	15.75	16.24	9.08	8.91	12.56	12.67	*[b]	*	10.07	10.03	2.59sg	2.48sg

[a] c = Cochran test outlier on insoluble dietary fiber; sg = Single Grubbs test outlier on insoluble dietary fiber; dg = Double Grubbs test outlier on insoluble dietary fiber.

[b] * = Laboratory reported no result for this sample.

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Table 3. Collaborative study data for total dietary fiber (%TDF by sum of IDF and SDF reported by laboratories for manual deionization)[a]

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1	28.90	27.82	9.87	9.80	27.05	28.32	11.78	12.02	21.03	20.22	17.12	17.35	22.90	23.04	17.25	17.81
2	27.93	29.32	10.61	10.04	24.01	23.56	12.26	12.21	19.62	19.53	15.76	16.85	20.17	20.74	17.64	17.97
3	31.11	33.23	11.75	11.23	25.31	29.92	12.32	13.29	23.31	24.63	17.89	15.60	23.34	25.59	19.95	19.93
4	30.37	31.04	11.85	12.61	30.24	31.45	12.33	12.20	21.93	22.57	14.60	17.69	25.46	26.81	20.48	19.50
5	30.33	31.59	11.19	11.56	27.54	28.24	11.80	13.56	21.86	22.00	22.86	17.61	25.16	25.56	19.78	18.91
6	32.66	35.66	9.91	11.70	29.18	27.51	13.38	13.50	21.24c	25.12c	43.04sg	43.07sg	29.42	29.63	27.07sg	29.79sg
7	28.96	28.90	10.91	10.68	24.20	24.52	10.35	10.55	19.07	19.19	14.41	14.80	23.07	23.73	18.23	18.41
8	28.87	28.75	9.39	9.95	25.97	29.51	10.06	10.23	24.15	24.94	19.63	21.29	34.48c	21.49c	22.79	20.47
9	28.28	28.75	9.65	9.18	24.14	24.71	10.88	10.20	21.29	20.72	14.06	14.29	21.10	21.14	17.41	16.64
10	31.64	32.15	9.88	10.77	28.98	25.01	11.50	11.45	22.49	21.70	16.56	18.52	29.14	28.43	17.28	17.12
11	31.30	31.42	11.32	11.73	30.30	29.56	11.88	12.70	22.26	22.34	15.26	15.93	23.42	24.22	19.29	18.45
12	27.07	26.45	9.67	10.10	21.97	21.00	12.02	13.31	19.24	18.31	13.95	13.41	20.52	17.38	16.22	16.43
13	28.20	28.04	10.07	10.00	26.60	27.06	11.99	12.41	21.21	22.66	16.04	16.75	24.41	24.16	17.36	18.43
14	29.95	30.66	10.19	9.60	26.88	26.91	11.69	11.53	20.25	20.65	13.84	14.04	21.56	22.99	17.66	16.92
15	28.12	29.59	9.24	9.02	23.34	23.54	10.25	10.01	20.63	20.67	*[b]	*	20.40	20.30	12.15sg	11.53sg

[a] c = Cochran test outlier on insoluble dietary fiber; sg = Single Grubbs test outlier on insoluble dietary fiber; dg = Double Grubbs test outlier on insoluble dietary fiber.

[b] * = Laboratory reported no result for this sample.

Collaborative Study Protocol

Eight food samples were selected for the collaborative study. Because the method under consideration incorporates resistant starch and nondigestible oligosaccharides into a more traditional dietary fiber methodology, the samples for this collaborative study were chosen to be challenging, i.e., with emphasis on quantitating products high in resistant starch (legumes, resistant starch ingredient, and whole grain products) and products with typical levels of nondigestible oligosaccharides. Methods designed to quantitate dietary fiber have been thoroughly studied and validated since 1980 (e.g., AOAC Methods **985.29**, **991.42**, **993.19**, **991.43**, etc.) matching the consensus Trowell et al. (6) dietary fiber definition of that time. Inclusion of components, such as resistant starch and nondigestible oligosaccharides, in the CODEX Alimentarius definition indicates that updated testing procedures must include the capability of accurately quantitating these components.

Sufficient quantities of sample for the collaborative study were collected and mixed. Samples were then ground to pass a 0.5 mm sieve and remixed post-grinding to ensure homogeneity. Moist samples were freeze-dried before grinding. After thorough mixing, samples were subdivided into polyethylene bottles and sealed. Samples, copies of the method, electronic report sheets, Excel-based calculators, and sample storage instructions, along with an adequate supply of enzymes and deionizing resins, were shipped to collaborating laboratories by express overnight shipment.

A total of 22 laboratories reported data for the collaborative study samples. One laboratory used high-pressure anion exchange chromatography with electrochemical detection instead of the prescribed LC method, and therefore, could not obtain accurate data for dietary fibers that are soluble in the alcohol-water solution. Another laboratory which had not previously performed dietary fiber analyses had difficulties with the alcohol-water soluble dietary fibers, and reported results that were significantly lower than those of all the other laboratories in all three categories, i.e., IDF, SDF, and TDF. The source of the discrepancy could not be determined. A third laboratory was only able to complete 10 of the 16 samples. Data from these three laboratories are not included in data tables or statistical analysis. In addition, four laboratories did not quantitate the IDF and SDF fractions of the TDF. The data from those four laboratories are included with the data from the five laboratories that quantitated IDF and SDF and measured TDF directly.

Statistical Treatment

Collaborating laboratory data were evaluated statistically according to AOAC protocols using AOAC-supplied software. Of the 118 valid pairs (236 data points) of assay results reported for IDF, Laboratories 1–5, 7, 9, and 11–15 had no statistical outliers, Laboratory 10 had one, Laboratory 8 had three, and

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Laboratory 6 had five, for a total of nine statistical outlier pairs overall. The raw data and statistically paired data from the blind duplicate results for IDF reported by the collaborating laboratories are shown in Tables 1 and **2011.25A**, respectively. Outliers, and the reason for outlier removal, are indicated in Table 1 footnotes. Of the 118 valid pairs of assay results reported for SDF, Laboratories 1, 2, 4, 7, and 9–14 had no statistical outliers, Laboratories 3, 5, 8, and 15 had one, and Laboratory 6 had two, for a total of six statistical outlier pairs overall. The raw data and statistically paired data from the blind duplicate results for SDF reported by the collaborating laboratories are shown in Tables 2 and **2011.25B**, respectively. Outliers, and the reason for outlier removal, are indicated in Table 2 footnotes. Of the 118 valid pairs of assay results reported for TDF, Laboratories 1–5, 7, and 9–14 had no statistical outliers, Laboratories 8 and 15 had one, and Laboratory 6 had three, for a total of five statistical outlier pairs overall. The raw data and statistically paired data from the blind duplicate results for TDF reported by the collaborating laboratories are shown in Tables 3 and **2011.25C**, respectively. Outliers, and the reason for outlier removal are indicated in Table 3 footnotes.

AOAC Official Method 2011.25 Insoluble, Soluble, and Total Dietary Fiber in Foods

Enzymatic-Gravimetric-Liquid Chromatography First Action 2011

[Applicable to plant material, foods, and food ingredients consistent with CODEX Alimentarius Commission Definition adopted in 2009 and modified slightly in 2010 (ALINORM 09/32/REP and ALINORM 10/33/REP, respectively), including naturally occurring, isolated, modified, and synthetic polymers meeting that definition.]

See Tables **2011.25A**–**H** for the results of the interlaboratory study supporting acceptance of the method.

A. Principle

A method is described for the measurement of insoluble, soluble, and total dietary fiber (IDF, SDF, and TDF, respectively), inclusive of the resistant starch (RS) and the water:alcohol soluble nondigestible oligosaccharides and polysaccharides of DP ≥3. The method combines the key attributes of AOAC *Official Methods of Analysis*SM **985.29** (and its extensions, **991.42** and **993.19**), **991.43**, **2001.03**, and **2002.02**. Duplicate test portions are incubated with pancreatic α-amylase and amyloglucosidase (AMG) for 16 h at 37°C in sealed 250 mL bottles while mixing with sufficient vigor to maintain continuous suspension. During this step, nonresistant starch is solubilized and hydrolyzed to glucose and maltose by the combined action of the two enzymes. The reaction is terminated by pH adjustment and temporary heating. Protein in the sample is then digested with protease. For the measurement of IDF, the digestate is filtered and the IDF is determined gravimetrically after correction for any protein or ash in the residue. For the measurement of the water soluble, but water:alcohol insoluble dietary fiber (SDFP), ethanol is added to the filtrate of the IDF; the precipitated SDFP is captured by filtration and determined gravimetrically after correction for any protein or ash in the precipitate. Nonprecipitable, water:alcohol soluble dietary fiber (SDFS) in the filtrate is recovered by concentrating the filtrate, deionizing through ion exchange resins, concentrating, and quantitating by LC, or, alternatively, by concentrating the filtrate and simultaneously deionizing and quantitating by LC.

B. Apparatus

(**a**) *Grinding mill*.—Centrifugal, with 12-tooth rotor and 0.5 mm sieve, or similar device. Alternatively, a cyclone mill can be used for small test laboratory samples provided the mill has sufficient air flow or other cooling to avoid overheating samples.

(**b**) *Digestion bottles*.—250 mL Fisherbrand® soda glass, wide-mouthed bottles with polyvinyl-lined cap (Cat. No. FB73219, equivalent: Cat. No. 02-911-916; Fisher USA, www.fisher.co.uk/supplies/fisherbrand.html) or 250 mL polypropylene bottles with polypropylene caps.

(**c**) *Fritted crucible*.—Gooch, fritted disk, Pyrex® 50 mL, pore size, coarse, ASTM 40–60 µm (Corning No. 32940-50C® or equivalent; http://www.labplanet.com/corning-crucible-gooch-high-c-50-ml-32940-50c.html). Prepare four for each sample as follows: Ash overnight at 525°C in muffle furnace. Cool furnace to 130°C before removing crucibles to minimize breakage. Remove any residual Celite and ash material by using a vacuum. Soak in 2% cleaning solution, **C**(**i**), at room temperature for 1 h. Rinse crucibles with water and deionized water. For final rinse, use 15 mL acetone and air dry. Add approximately 1.0 g Celite to dried crucibles and dry at 130°C to constant weight. Cool crucible in desiccators for approximately 1 h and record mass of crucible containing Celite.

(**d**) *Filtering flask*.—Heavy-walled, 1 L with side arm.

(**e**) *Rubber ring adaptors*.—For use to join crucibles with filtering flasks.

(**f**) *Vacuum source*.—Vacuum pump or aspirator with regulator capable of regulating vacuum.

(**g**) *Water bath(s)*.—Rotary motion, large-capacity (20–24 L) with covers; capable of maintaining temperature of 37 ± 1°C and 60 ± 1°C; equipped with automatic timers for on-off operation or equivalent (e.g., Grant® OLS 200 shaking incubation bath; http://www.bioresearchonline.com/product.mvc/Grant-OLS200-Orbital-Shaking-Water-Bath-0001). Ensure that shaking action/sample agitation in water bath used is sufficient to maintain sample solids in suspension and no residue build-up or rings of sample material form in the digestion bottle during the enzymatic digestions. A linear motion (back and forth shaker) can be used if the bottles are placed at 45° to ensure adequate agitation (if the bottles are vertical or horizontal there will not be sufficient agitation to ensure that the sample remains suspended). Alternatively, a 2 mag Mixdrive 15 submersible magnetic stirrer apparatus (http://www.2mag.de/english/stirrer/multiple/stirrer_multiple_04_mixdrive6_15.html) can be used in a water bath maintained at 37 ± 1°C with a Julabo circulating heater (http://www.julabo.de/). Samples are stirred in digestion bottles, (**b**), with a 7 × 30 mm stir bar at ~170 rpm (Figure **2011.25A**).

(**h**) *Balance*.—0.1 mg readability, accuracy, and precision.

(**i**) *Ovens*.—Two, mechanical convection, set at 105 ± 2°C and 130 ± 3°C, respectively.

(**j**) *Timer*.

(**k**) *Desiccator*.—Airtight, with SiO_2 or equivalent desiccant. Desiccant dried biweekly overnight in 130°C oven.

(**l**) *pH meter*.

(**m**) *Pipettors and tips*.—Positive displacement 50–200 µL and 5 mL capacity.

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Table 2011.25A. Interlaboratory study results for IDF

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	14	15	15	15	13	13	12	13
Mean, %	25.74	4.79	9.23	1.05	8.26	11.63	11.52	9.90
S_r	0.52	0.18	0.71	0.13	0.19	0.48	0.33	0.29
S_R	1.23	0.47	2.24	0.42	0.48	1.12	1.21	0.77
RSD_r	2.03	3.76	7.67	12.25	2.26	4.11	2.87	2.98
RSD_R	4.76	9.82	24.35	39.64	5.81	9.65	10.53	7.80
HorRat	1.94	3.11	8.51	9.98	2.00	3.49	3.80	2.75

(**n**) *Dispensers*.—(*1*) 15 ± 0.5 mL for 78% EtOH or industrial methylated spirits (IMS), 95% ethanol or IMS, and acetone. (*2*) 40 ± 0.5 mL for buffer.

(**o**) *Cylinder*.—Graduated, 500 mL.

(**p**) *Magnetic stirrers and stirring bars*.

(**q**) *Rubber spatulas*.

(**r**) *Muffle furnace*.—525 ± 5°C.

(**s**) *Polypropylene columns*.—(If manual deionization is carried out.) For the interlaboratory study, Alltech SPE cartridges [Extract-Clean™ reservoirs (15 mL), Cat. No. 210315; http://www.discoverysciences.com/uploadedFiles/Home/SPE_Filtration_ExtractNUltraClean_p302to306.pdf] were used. Ten grams of cation or anion exchange resins were added to each column. The cartridges were connected with an Alltech Syringe Adapter™ (Cat. No. 210705), with cation exchange resin [Ambersep 200C (H^+)] always at the top. Flow was controlled by pump or with an Alltech One-Way Stopcock (Cat. No. 211524; Figure **2011.25B**). A method for resin regeneration was provided, but reuse of resins is no longer recommended since resin cost is minimal. However, Extract-Clean reservoirs and Syringe Adapters subsequently received were unsatisfactory. There was some leakage of liquid due to incomplete sealing. Subsequently, sample deionization has been achieved using a mixed bed resin of approximately 4 g cation [Ambersep 200C (H^+)] and 4 g anion [Amberlite FPA53 (OH^-)] in a Bio-Rad Laboratories Econo-Pac™ disposable chromatography column (Cat. No. 732-1010; http://www.bio-rad.com/webroot/web/pdf/psd/literature/Bulletin_9103.pdf) with an Alltech One-Way Stopcock (Cat. No. 211524; Figure **2011.25B**).

(**t**) *Cation and anion exchange guard columns*.—(If online simultaneous deionization and quantitation are carried out.) Cation and anion exchange guard columns, H^+ and CO_2^{3-} forms, respectively (Bio-Rad Laboratories, Cat. No. 125-0118, includes one cation and one anion cartridge), with guard column holder (Bio-Rad Laboratories, Cat. No. 125-0139) to hold the two guard column cartridges in series, cation cartridge preceding anion cartridge.

(**u**) *Liquid chromatograph*.—(*1*) *Pump*.—Capable of precision metering at 0.5 mL/min at up to 10 000 psi. Arrangement for HPLC of manually desalted samples on Waters Sugar-Pak® column (http://www.waters.com/waters/support.htm?locale=?&cid=513789&lid=10069619) is shown in Figure **2011.25C**. [If using Sugar-Pak column and simultaneous online deionization and quantitation, a second pump capable of precision metering at 0.05 mL/min at up to 10 000 psi is needed to add disodium calcium ethylenediaminetetracetate ($Na_2CaEDTA$) via zero dead volume t-fitting to the line between the deionization cartridges and the quantitation column (*see* Figure **2011.25D**).]

(*2*) *Filled loop injector*.—Equipped with 50 µL injection loop.

(*3*) *LC column*.—Waters Sugar-Pak 6.5 × 300 mm (Part No. WAT085188) or equivalent preceded by Waters Guard Pak LC precolumn inserts (Part No. WAT015209) or equivalent.

(*4*) *Refractive index (RI) detector*.—Capable of being maintained at 50°C.

(*5*) *Oven*.—To maintain a column temperature of 90°C.

(**v**) *Data integrator or computer*.—For peak area measurement.

(**w**) *Filters for disposable syringe*.—Polyvinylidene fluoride 0.45 µm, 13 or 25 mm.

(**x**) *Filters for water*.—Polyvinylidene fluoride, 0.45 µm, 47 mm.

(**y**) *Filter apparatus*.—To hold 47 mm, 0.45 µm filter, (**x**); to filter larger volumes of water.

(**z**) *Syringes*.—10 mL, disposable, plastic.

(**aa**) *Syringes*.—Hamilton® 100 µL, 710SNR syringe.

Table 2011.25B. Interlaboratory study results data for SDF

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	14	14	15	15	14	13	15	13
Mean, %	3.80	5.58	17.32	10.74	12.94	4.56	11.29	8.30
S_r	0.44	0.28	1.03	0.46	0.50	0.83	0.71	0.51
S_R	0.99	0.85	1.30	1.12	1.52	1.66	1.61	1.33
RSD_r	11.71	5.10	5.97	4.33	3.86	18.10	6.29	6.18
RSD_R	26.20	15.20	7.53	10.46	11.73	36.40	14.25	16.07
HorRat	8.01	4.92	2.89	3.74	4.31	11.44	5.13	5.53

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Table 2011.25C. Interlaboratory study results for TDF

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	15	15	15	15	14	13	14	13
Mean, %	29.90	10.45	26.55	11.79	21.37	16.39	23.71	18.40
S_r	0.86	0.47	1.39	0.49	0.52	1.41	0.87	0.64
S_R	2.05	0.95	2.74	1.10	1.72	2.37	3.14	1.56
RSD_r	2.88	4.51	5.25	4.17	2.43	8.60	3.65	3.47
RSD_R	6.85	9.11	10.31	9.30	8.04	14.48	13.23	8.47
HorRat	2.85	3.24	4.22	3.37	3.19	5.51	5.33	3.28

(**bb**) *Rotary evaporator*.—Heidolph Laborota® 4000 or equivalent (http://www.lasallescientific.com/products/heidolph/BAL_LARO4000_Nov06_en1.pdf).

(**cc**) *Thermometer*.—Capable of measuring to 110°C.

(**dd**) *Freeze drying apparatus*.

C. Reagents

(**a**) *Ethanol, 95% (v/v) or alternatively use IMS*.—Specifications for preparation or commercial purchase: ethanol 84.83 (w%), 85.95 (v%); water 5.66 (w%), 4.52 (v%); 2-propanol 4.91 (w%), 5.00 (v%); methanol 4.60 (w%), 4.52 (v%). Alternatively, IMS can be prepared by mixing five volumes of 2-propanol with 95 volumes of denatured ethanol formula SDA-3A (SDA-3A alcohol consists of 100 volumes of 95% ethanol combined with five volumes of methanol).

(**b**) *Ethanol (or IMS), 78% (v/v)*.—Place 207 mL water into 1 L volumetric flask. Dilute to volume with 95% ethanol or IMS. Mix.

(**c**) *Acetone*.—Reagent grade.

(**d**) *Stock AMG solution*.—3300 Units/mL in 50% (v/v) glycerol. Solution is viscous; for dispensing use positive displacement dispenser. AMG solution is stable for up to 5 years when stored at 4°C. (*Note*: One unit of enzyme activity is amount of enzyme required to release 1 micromole glucose from soluble starch per minute at 40°C and pH 4.5.) AMG solution should be essentially devoid of β-glucanase and β-xylanase and free of detectable levels of free glucose (*see* **E**).

(**e**) *Pancreatic α-amylase (50 Units/mL)/AMG (3.4 Units/mL)*.—Immediately before use, dissolve 0.10 g purified porcine pancreatic α-amylase (~130 000 Units/g; Ceralpha method; AOAC Method **2002.01**) in 290 mL sodium maleate buffer (50 mM, pH 6.0 plus 2 mM $CaCl_2$ and 0.02% sodium azide) and stir for 5 min. Add 0.3 mL AMG stock solution (*see* **E**).

(**f**) *Protease (50 mg/mL; ~350 Tyrosine Units/mL) in 50% (v/v) glycerol*.—Solution is viscous; for dispensing use positive displacement dispenser. Protease must be devoid of α-amylase and essentially devoid of β-glucanase and β-xylanase (*see* **E**). Use as supplied.

(**g**) *Deionized water*.

(**h**) *Celite*.—Acid-washed, pre-ashed (Megazyme G-CEL100 or G-CEL500, Megazyme International Ireland Ltd., Bray, County Wicklow, Ireland, http://www.megazyme.com/Dynamic.aspx?control=CSViewProduct&categoryName=GeneralChemicals&productId=G-CEL100; or Sigma-Aldrich Cat. No. C8656, http://www.sigmaaldrich.com/catalog/ProductDetail.do?D7=0&N5=SEARCH_CONCAT_PNO%7CBRAND_KEY&N4=C8656%7CSIGMA&N25=0&QS=ON&F=SPEC).

(**i**) *Cleaning solution*.—Micro-90® (International Products Corp., Burlington, NJ). Make a 2% solution (w/v) with deionized water.

(**j**) *Sodium maleate buffer*.—50 mM, pH 6.0 plus 2 mM $CaCl_2$ and 0.02% sodium azide. Dissolve 11.6 g maleic acid in 1600 mL distilled water and adjust pH to 6.0 with 4 M (160 g/L) NaOH solution. Add 0.6 g calcium chloride dihydrate and 0.4 g sodium azide [*Note:* Do not add the sodium azide until the pH has been adjusted. Acidification of sodium azide releases a poisonous gas. Handle sodium azide and maleic acid with caution only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.] and adjust the volume to 2 L. Stable for >1 year at 4°C.

(**k**) *Trizma base*.—0.75 M (Sigma Cat. No. T-1503). Add 90.8 g Trizma base to approximately 800 mL distilled water and dissolve. Adjust volume to 1 L. Stable for >1 year at room temperature.

Table 2011.25D. Interlaboratory study results for TDF measured directly

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	9	8	9	9	9	7	9	9
Mean, %	28.08	9.97	24.60	11.00	21.21	16.47	21.27	16.47
S_r	1.26	0.41	0.70	0.39	1.11	0.72	1.19	1.16
S_R	1.31	1.24	1.95	2.54	2.30	5.27	2.10	2.58
RSD_r	4.50	4.16	2.83	3.57	5.23	4.38	5.61	7.04
RSD_R	4.67	12.41	7.95	23.11	10.86	32.00	9.89	15.66
HorRat	1.93	4.39	3.22	8.29	4.30	12.20	3.92	5.97

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Table 2011.25E. Interlaboratory study results for SDF as measured using online deionization

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	6	6	6	6	6	6	5	6
Mean, %	3.45	4.87	16.29	8.60	11.93	2.95	10.98	7.11
S_r	0.34	0.29	0.58	0.49	0.24	0.16	0.24	0.26
S_R	0.64	0.49	1.24	1.99	0.83	0.49	0.40	1.07
RSD_r	9.98	5.94	3.55	5.73	2.00	5.28	2.18	3.64
RSD_R	18.63	10.12	7.60	23.12	6.96	16.73	3.64	15.08
HorRat	5.61	3.21	2.89	7.99	2.53	4.92	1.31	5.07

(**l**) *Acetic acid solution, 2 M*.—Add 115 mL glacial acetic acid (Fluka 45731) to a 1 L volumetric flask. Dilute to 1 L with distilled water.

(**m**) *pH standards*.—Buffer solutions at pH 4.0, 7.0, and 10.0.

(**n**) *Fructooligosaccharides*.—From chicory (Sigma Aldrich Cat. No. F8052).

(**o**) *LC retention time standard*.—Standard source having the distribution of oligosaccharides (DP ≥3) corn syrup solids (DE 16; Matsutani Chemical Industry Co., Ltd., Itami City, Hyogo, Japan; http://www.matsutani.com/productsframeset.html, maltodextrin TK-16; or Sigma Aldrich Cat. No. 419699), analyzed by LC. Dissolve 2.0 g portion of oligosaccharide mixture and 0.50 g maltose in deionized water and transfer to 100 mL volumetric flask. [Alternatively, dissolve 2.5 g LC retention time standard mix from Integrated Total Dietary Fiber Assay Kit (K-INTDF; Megazyme International Ireland Ltd), which contains maltodextrins with added maltose.] Pipet 10 mL of stock internal standard solution, (**q**). Bring to volume with 0.02% sodium azide solution, (**u**). Transfer solution to 100 mL polypropylene bottle. Stable at room temperature for 1 year.

(**p**) *Manual deionization*.—(*1*) *Original method*.—*Column 1*.—10 g Ambersep 200 (H^+) resin, ion exchange capacity (R-H exchange capacity data supplied by manufacturer) 1.6 meq/mL (min), or equivalent (Rohm and Haas, France S.A.S.; *see* also Megazyme G-AMBH). *Column 2*.—10 g Amberlite FPA53 (OH^-) resin, 1.6 meq/mL (min) or equivalent (Rohm and Haas, France S.A.S.; *see* also Megazyme G-AMBOH) packed in separate columns [Extract-Clean reservoirs (15 mL), **B**(**s**)] for analysis of each test portion. Before connecting the columns, wash the resin in each column with sufficient H_2O to obtain a pH value of 4–7 for column 1 and 7–8 for column 2.

(*2*) *Updated method*.—Mix approximately 4 g Ambersep 200 (H^+) resin or equivalent, 1.6 meq/mL (min) or equivalent (Rohm and Haas, France S.A.S.; *see* also Megazyme G-AMBH) with approximately 4 g Amberlite FPA53 (OH^-) resin, 1.6 meq/mL (min) or equivalent (Rohm and Haas, France S.A.S.; *see* also Megazyme G-AMBOH), ion exchange capacity (R-OH exchange capacity data supplied by manufacturer), in a 100 mL beaker and slurry with a minimum volume of distilled water. Pour this mixture into a Bio-Rad Econo-Pac disposable chromatography column (Cat. No. 732-1010) fitted with an Alltech One-Way Stopcock (Cat. No. 211524). Place a small wad of cotton wool into the top of the column and wash the resin with 20 mL distilled water. The column is then ready for use. If alternative resins are used, determine that carbohydrates are not retained by the resin by preparing a test solution consisting of 1 mL of 100 mg/mL internal standard and 2.5 mL of 10 mg/mL fructooligosaccharides diluted to 10 mL. Add 2 mL of this solution to the mixed bed deionization resin (approximately 8 g) as per **I**(**e**). Recovery of the internal standards and fructooligosaccharides should match that of the solution injected directly onto the LC.

(**q**) *D-sorbitol (stock internal standard)*.—100 mg/mL. [*Note*: Sorbitol is present in some foods and food products, such as plums (prunes), pears, apples, and Polydextrose® and may also be added to foods. If sorbitol is present, use of the internal standard will result in artificially low SDFS quantity. In these cases, the external standard method of calculation should be used. Addition of the internal standard is still recommended to use as a retention time reference.] Weigh 100 g dry D-sorbitol into a beaker, dissolve in water, transfer to a 1 L volumetric flask with water, and dilute to volume. Transfer to a polypropylene bottle and add 0.2 g sodium azide as a preservative. (*Note*:

Table 2011.25F. Interlaboratory study results for TDF (sum of IDF and SDF) as measured using online deionization

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	6	6	6	4	6	6	5	6
Mean, %	29.65	9.95	26.96	11.21	20.47	15.06	23.04	17.43
S_r	0.73	0.36	0.73	0.27	0.32	0.43	0.25	0.24
S_R	1.29	0.66	2.41	0.37	0.90	1.16	1.71	0.77
RSD_r	2.47	3.58	2.70	2.42	1.58	2.87	1.09	1.36
RSD_R	4.33	6.62	8.95	3.33	4.40	7.67	7.40	4.41
HorRat	1.81	2.34	3.67	1.20	1.73	2.88	2.97	1.69

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Table 2011.25G. Interlaboratory study results for SDF calculated using external standard approach

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	11	12	12	12	12	11	11	10
Mean, %	3.94	5.49	16.04	10.06	12.52	3.92	11.34	8.09
S_r	0.45	0.54	1.48	0.81	0.67	0.69	0.44	0.63
S_R	1.00	1.11	2.02	1.81	2.10	1.20	0.95	1.99
RSD_r	11.42	9.92	9.21	8.06	5.37	17.58	3.93	7.81
RSD_R	25.40	20.20	12.62	17.95	16.74	30.58	8.39	24.58
HorRat	7.80	6.53	4.79	6.35	6.12	9.39	3.02	8.42

Handle sodium azide with caution only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.) Seal well. Stable at room temperature for >1 year.

(**r**) *D-glucose LC standards (5, 10, 20 mg/mL)*.—Accurately weigh 0.5, 1.0, and 2.0 g portions of high-purity (>99.5%) D-glucose (Sigma Chemical Co.; http://www.sigmaaldrich.com/ireland.html) and transfer to three separate 100 mL volumetric flasks respectively. To each flask pipet 10 mL of internal standard, (**q**). Bring to volume with 0.02% sodium azide solution, (**u**). Transfer solutions to 100 mL polypropylene bottles. Stable at room temperature for 1 year.

(**s**) *Disodium calcium ethylenediaminetetracetate ($Na_2CaEDTA$; 50 mg/L; if using Sugar-Pak LC column and manual deionization)*.—Dissolve 50 mg $Na_2CaEDTA$ in about 500 mL deionized water in a 1 L volumetric flask; dilute to volume.

(**t**) *$Na_2CaEDTA$ (500 mg/L; if using Sugar-Pak LC column and simultaneous online deionization and quantitation)*.—Dissolve 500 mg $Na_2CaEDTA$ in about 500 mL deionized water in a 1 L volumetric flask; dilute to volume.

(**u**) *Sodium azide solution (0.02%, w/v)*.—Add 0.2 g sodium azide to 1 L deionized water and dissolve by stirring. (*Note:* Handle sodium azide with caution only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.) Stable at room temperature for >1 year.

Items (**d**)–(**f**) and (**n**) are supplied in K-INTDF (Megazyme International Ireland Ltd), but preparations of reagents and buffers which meet the criteria as specified in the method above may also be used.

D. Preparation of Test Samples

Collect and prepare samples as intended to be eaten, i.e., baking mixes should be prepared and baked, pasta should be cooked, etc. Defat per AOAC **985.29** if >10% fat. For high-moisture samples (>25%), it may be desirable to freeze dry. Grind ca 50 g in a grinding mill, **B**(**a**), to pass a 0.5 mm sieve. Transfer all material to a wide-mouthed plastic jar, seal, and mix well by shaking and inversion. Store jar in the presence of a desiccant.

E. Enzyme Purity

To ensure absence of undesirable enzymatic activities and effectiveness of desirable enzymatic activities, run standards listed in Table **985.29** each time enzyme lot changes or at a maximum 6 month interval.

F. Enzymatic Digestion of Sample

(**a**) *Blanks*.—With each assay, run blanks along with samples to measure any contribution from reagents to residue.

(**b**) *Samples*.—(*1*) Weigh duplicate 1.000 ± 0.005 g samples accurately into 250 mL Fisherbrand bottles, **B**(**b**).

(*2*) *Addition of enzymes*.—Wet sample with 1.0 mL ethanol (or IMS) and add 40 mL pancreatic α-amylase/AMG mixture, **C**(**e**), to each bottle. Cap the bottles. Transfer bottles to the orbital or linear shaking incubation bath and secure bottles in place with the springs in the shaker frame. Alternatively, use the submersible magnetic stirrer apparatus, **B**(**g**).

(*3*) *Incubation with pancreatic α-amylase/AMG*.—Incubate the reaction solutions at 150 revolutions/min (orbital motion) or at the relevant rate to ensure suspension in a reciprocal shaker at 37°C for 16 h ± 10 min (e.g., 5:00 pm to 9:00 am). Alternatively, stir the contents of the Fisherbrand bottles, **B**(**b**), on a 2 mag Mixdrive 15 stirring device (with 7 × 30 mm stirrer bar) submerged in a water bath at 37°C for 16 h ± 10 min (Figure **2011.25A**).

(*4*) *Adjustment of pH to approximately 8.2 (pH 7.9–8.4), inactivation of α-amylase and AMG*.—After 16 h, remove sample bottles from the shaking water bath and immediately

Table 2011.25H. Interlaboratory study results for TDF calculated using external standard approach

Sample/parameter	A	B	C	D	E	F	G	H
No. of labs	12	12	12	12	12	10	12	10
Mean, %	30.05	10.35	25.52	11.20	20.93	15.58	23.52	18.05
S_r	1.05	0.59	1.52	0.85	0.96	0.85	0.77	0.60
S_R	1.95	1.13	2.08	1.71	1.97	1.27	2.61	1.55
RSD_r	3.49	5.67	5.95	7.57	4.61	5.49	3.26	3.32
RSD_R	6.47	10.90	8.13	15.26	9.41	8.18	11.08	8.61
HorRat	2.70	3.87	3.31	5.49	3.72	3.09	4.46	3.33

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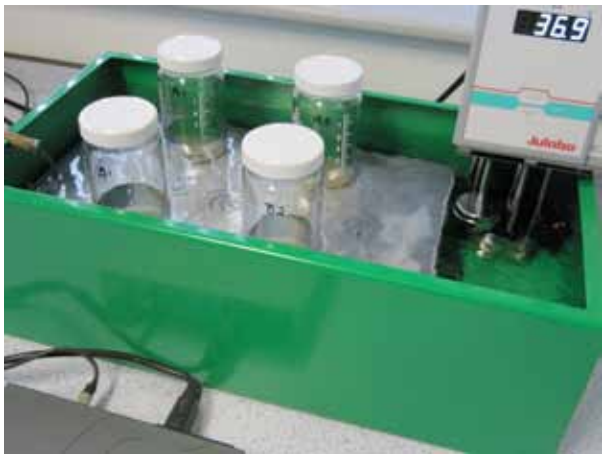

Figure 2011.25A. Typical arrangement for sample mixing using a 2 mag Mixdrive 15 submersible magnetic stirrer in a custom made water bath.

add 3.0 mL of 0.75 M Trizma base solution to terminate the starch digestion reactions. (At the same time, if only one shaker bath is available, increase the temperature of the shaking incubation bath to 60°C in readiness for the protease incubation step.) Slightly loosen the caps of the sample bottles and place the bottles in a water bath (nonshaking) at 95–100°C, and incubate for 20 min with occasional shaking (by hand). Using a thermometer, ensure that the final temperature of the bottle contents is >90°C (checking of just one bottle is adequate).

(*5*) *Cool*.—Remove all sample bottles from the hot water bath (use appropriate gloves) and cool to approximately 60°C.

(*6*) *Protease treatment*.—Add 0.1 mL protease solution, **C**(**f**), using a positive displacement dispenser (solution is viscous). Incubate at 60°C for 30 min.

(*7*) *pH adjustment*.—Add 4.0 mL of 2 M acetic acid, **C**(**l**), to each bottle and mix. This gives a final pH of approximately 4.3.

(*8*) *Addition of internal standard*.—To each sample, add 1 mL of 100 mg/mL internal standard solution, **C**(**q**).

(*9*) Proceed to **G**(**a**) for determination of IDF when IDF and SDF values are desired. Proceed to **O**(**a**) when TDF is desired without fractionation to IDF and SDF.

G. Determination of IDF

(**a**) *Filtration setup*.—Tare crucible containing Celite to nearest 0.1 mg. Wet and redistribute the bed of Celite in the crucible, using 15 mL of 78% (v/v) EtOH (or IMS), **C**(**b**), from wash bottle. Apply suction to crucible to draw Celite onto fritted glass as an even mat. Discard this filtrate.

(**b**) *Filtration*.—Using vacuum, filter enzyme digest, **F**(**b** (*9*), through crucible. With the aid of a wash bottle use 21.9 mL or less of deionized water (warmed to 60°C) to quantitatively transfer remaining particles to crucible and rinse residue. Retain combined filtrate and transfer water and set aside for determination of water SDF, **H**(**a**).

(**c**) *Wash*.—Using vacuum, wash residue two times each with 15 mL portions of 78% ethanol, 95% ethanol, and acetone. (*Note*: Delay in washing IDF residues with 78% ethanol, 95% ethanol, and acetone may cause inflated IDF values.) Discard these washings.

(**d**) Dry crucibles containing residue overnight in 105°C oven.

(**e**) Cool crucible in desiccator for approximately 1 h. Weigh crucible containing IDF residue and Celite to nearest 0.1 mg. To obtain residue mass, subtract tare weight (i.e., weight of dried crucible and Celite).

(**f**) *Protein and ash determination*.—Residue from one crucible of the duplicates is analyzed for protein, and the second residue of the duplicates is analyzed for ash. Perform protein analysis on residue using Kjeldahl or combustion methods. (Caution should be exercised when using a combustion analyzer for protein in the residue. Celite volatilized from the sample can clog the transfer lines of the unit.) Use 6.25 factor for all cases to calculate mg of protein. For ash analysis, incinerate the second residue for 5 h at 525°C. Cool in desiccator and weigh to nearest 0.1 mg. Subtract crucible and Celite weight to determine ash.

(**g**) Proceed to determination of water soluble dietary fiber, **H**(**a**).

H. Determination of Water Soluble Dietary Fiber

(**a**) *Precipitation of water soluble SDFP*.—To each sample filtrate, add any remaining water from the IDF transfer and rinse, **G**(**b**), or additional water if necessary to bring the total volume to exactly 70 mL, then add 279 mL (measured at room temperature) of 95% (v/v) EtOH or IMS, **C**(**a**), preheated to 60°C and mix thoroughly. Allow the SDFP precipitate to form at room temperature for 60 min.

(**b**) *Filtration setup*.—Tare crucible containing Celite to nearest 0.1 mg. Wet and redistribute the bed of Celite in the crucible, using 15 mL of 78% (v/v) EtOH or IMS, **C**(**b**), from wash bottle. Apply suction to crucible to draw Celite onto fritted glass as an even mat. Discard this filtrate.

(**c**) *Filtration*.—Using vacuum, filter precipitated SDFP, **H**(**a**), from supernatant through crucible. Using a wash bottle with 78% (v/v) EtOH or IMS, **C**(**b**), quantitatively transfer all remaining particles to crucible. Retain filtrate and washings, (**c**) and (**d**), respectively, for determination of SDFS, **I**(**d**).

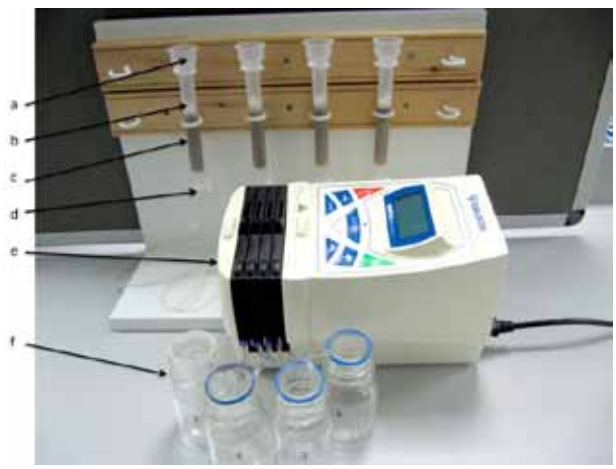

Figure 2011.25B. Column and pump arrangement for deionization of sample extracts: (a) Bio-Rad Econo-Pac® disposable chromatography column (Cat. No. 732-1010); (b) cotton wool plug; (c) a mixture of approximately 4 g Ambersep 200(H^+) resin and approximately 4 g Amberlite FPA53 (OH^-) resin; (d) stopcock, one-way tap; (e) Gilson Minipuls® Evolution pump; (f) Duran bottle collection containers.

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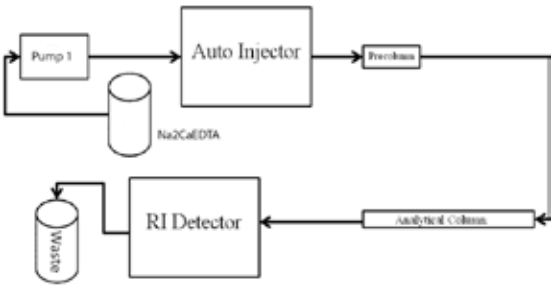

Figure 2011.25C. Apparatus assembly of liquid chromatography system for Sugar-Pak analytical column—manual deionization.

(**d**) *Wash*.—Using a vacuum, wash residue successively with two 15 mL portions of the following: 78% (v/v) EtOH or IMS; 95% (v/v) EtOH or IMS; acetone. Combine with filtrate of previous step.

(**e**) Dry crucibles containing residue overnight in 105°C oven.

(**f**) Cool crucible in desiccator for approximately 1 h. Weigh crucible containing dietary fiber residue and Celite to nearest 0.1 mg. To obtain residue mass, subtract tare weight, i.e., weight of dried crucible and Celite.

(**g**) *Protein and ash determination*.—Residue from one crucible of the duplicates is analyzed for protein, and the second residue of the duplicates is analyzed for ash. Perform protein analysis on residue using Kjeldahl or combustion methods. (Caution should be exercised when using a combustion analyzer for protein in the residue. Celite volatilized from the sample can clog the transfer lines of the unit.) Use 6.25 factor for all cases to calculate mg of protein. For ash analysis, incinerate the second residue for 5 h at 525°C. Cool in desiccator and weigh to nearest 0.1 mg. Subtract crucible and Celite weight to determine ash.

(**h**) Proceed to **I**(**d**).

I. Determination of SDFS

Chromatography of maltodextrins, maltose, glucose, sorbitol, and glycerol on a Sugar-Pak column is shown in Figure **2011.25E** and of water:alcohol soluble dietary fiber (WASDF or SDFS), glucose, and sorbitol in Figure **2011.25F**. *Note*: Proper deionization is an essential part of obtaining quality chromatographic data on SDFS. For manual deionization, be sure to obtain familiarity regarding the appearance of salt peaks in the SDFS chromatograms which may interfere with dietary fiber determination. In evaluating the column containing approximately 8 g of mixed bed resin, dissolve 10 mg sodium chloride into 9 mL deionized water and add 1 mL of 100 mg/mL LC internal standard, **C**(**q**), and proceed to **I**(**e**) at "transfer 2 mL to the top of the column of ...". The liquid chromatogram of this solution should show no peaks in the time range corresponding to carbohydrates of DP ≥3. For simultaneous online deionization and quantitation by LC, *see* Figure **2011.25F** regarding the change in chromatography which occurs as the ion exchange capacity of the cartridges is exceeded.

(**a**) *LC system setup*.—Set up the LC system choosing between the setups shown schematically in Figures **2011.25C** and **D** depending upon whether manual or online deionization is used.

Sugar-Pak column operating conditions are:

(*1*) *Temperature*.—90°C.

(*2*) *Manual deionization-mobile phase*.—Distilled water plus $Na_2CaEDTA$ (50 mg/L); flow rate, 0.5 mL/min.

(*3*) *Simultaneous online deionization and quantitation*.—Mobile phase for pump 1, distilled water; flow rate, 0.45 mL/min. Mobile phase for pump 2, $Na_2CaEDTA$ (500 mg/L); flow rate, 0.05 mL/min.

(*4*) System must clearly separate maltose from higher malto-oligosaccharides (Figure **2011.25E**).

(*5*) *Run time*.—30 min to ensure that all materials from the injection are cleared from the column prior to the next injection.

(*6*) *RI detector at 50°C*.—*Note*: For samples containing significant amounts of short-chain polyfructose, it is recommended that chromatography of the SDFS be carried out on 2-size exclusion LC columns (TSK-Gel 30 cm × 7.8 mm) connected in series (Sigma Aldrich Part No. 808020) using conditions described in AOAC *Official Method*SM **2009.01**.

(**b**) *Calibrate the area of chromatogram to be measured for SDFS*.—Use a 100 µL LC syringe, **B**(**aa**), to fill the 50 µL injection loop with retention time standard, **C**(**o**). Determine demarcation point between DP 2 and 3 oligosaccharides (disaccharides sucrose and maltose versus higher oligosaccharides). *See* Figures **2011.25E** and **F**.

(**c**) *Determine response factor (Rf) for D-glucose*.—Since D-glucose provides an LC RI response equivalent to the Rf of the nondigestible polysaccharides that make up SDFS, the LC is calibrated using D-glucose, and the Rf is used for determining the mass of SDFS. Use a 100 µL LC syringe, **B**(**aa**), to fill the 50 µL injection loop for each internal standard/D-glucose solution, **C**(**r**).

(*1*) *Internal standard method*.—Obtain the values for the peak areas of D-glucose and internal standard from the chromatograms. The reciprocal of the slope obtained by comparing the ratio of peak area of D-glucose/peak area of internal standard (*y*-axis) to the ratio of the mass of D-glucose/mass of internal standard (*x*-axis) is the "response factor." Determine the average Rf (typically 0.91 relative to D-sorbitol internal standard).

$$\text{Response factor (Rf)} = (\text{PA-IS})/(\text{PA-Glu}) \times (\text{Wt-Glu}/\text{Wt-IS})$$

where PA-IS = peak area internal standard (D-sorbitol); PA-Glu = peak area D-glucose; Wt-Glu = mass of D-glucose (mg)

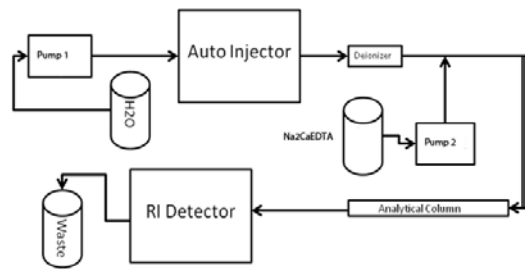

Figure 2011.25D. Apparatus assembly of liquid chromatography system for Sugar-Pak analytical column—simultaneous online deionization and quantitation.

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contained in 1 mL of standard (5, 10, or 20 mg); Wt-IS = mass of internal standard (mg) contained in 1 mL of standard (10 mg D-sorbitol).

(*2*) *External standard method*.—Obtain the values for the peak areas of D-glucose from the three chromatograms. Determine the average Rf:

$$Rf = (Wt\text{-}Glu)/(PA\text{-}Glu)$$

where Wt-Glu = mass (mg) of D-glucose contained in 1 mL of standard (5, 10, or 20 mg); PA-Glu = peak area D-glucose.

(**d**) *Filtrate recovery, deionization, and LC analysis*.—(Set aside the filtrate from one of the sample duplicates, **H**(**c**), to use in case of spills or take it through the procedure if duplicate SDFS data is desired.) Transfer filtrate(s), **H**(**c**), of the other sample duplicate to a 1 L evaporator flask and concentrate with a rotary evaporator to dryness at 50°C. Dissolve the residue in 10 mL deionized water, and transfer to a sealable container and store at 4°C if planning to store overnight before deionization. (In subsequent work, half of the filtrate was concentrated to dryness and redissolved in 5 mL water. This, as expected gave the same results, but halved the time for rotary evaporation.) For manual deionization, proceed to **I**(**e**). For simultaneous online LC deionization and quantitation, proceed to **I**(**f**).

(**e**) *Deionization of sample*.—In the interlaboratory study, manual deionization was performed using separate columns of Ambersep 200 (H^+) resin and Amberlite FPA53 (OH^-) resins (10 g of each) in Alltech Extract-Clean reservoirs (15 mL) connected with an Alltech Syringe Adapter (Cat. No. 210705). However, with subsequent lots of columns and adaptors, a perfect connection was not obtained and solutions were lost, so it was decided to use mixed bed resin [Ambersep 200 (H^+) and Amberlite FPA53 (OH^-)] (4 g of each) in a Bio-Rad Econo-Pac disposable chromatographic column (Cat. No. 732-1010) and to reduce sample size by the same percentage (40%; i.e., from 5 to 2 mL). Since the ratio of sample and resins remained the same, identical results were obtained. Weigh approximately 4 g Ambersep 200 (H^+) and approximately 4 g Amberlite FPA53 (OH^-) resins into a 100 mL beaker and slurry with a minimum volume of distilled water. Apply this slurry to a Bio-Rad Econo-Pac column and use distilled water from a wash bottle to transfer all resin to the column. Add a small cotton wool plug to the top of the resin and wash the column contents with approximately 20 mL deionized water. Connect the column to a Gilson Minipuls® Evolution pump (http://www.gilson.com/en/Pipette/Products/53.229, or similar). Centrifuge or filter the sample solution and transfer 2 mL to the top of the column of mixed bed resin and allow the column to elute at approximately 1 mL/min. Collect the eluate in a 250 mL round-bottom, rotary evaporator flask. When the liquid level has dropped to the level of the cotton wool plug, rinse the sample in with 2 mL distilled water and allow this to percolate into the resin collecting the eluate. Then add approximately 20 mL distilled water to the top of the column and allow this to elute into the 250 mL round-bottom flask at approximately 1 mL/min. Collect the eluate until the liquid level reaches the top of the resin in the column. Evaporate the solution just to dryness at 50°C. Dissolve the sample to bring it up to 2 mL of volume with distilled water by swirling the flask by hand for approximately 2 min. Transfer the sample into a polypropylene storage tube using a Pasteur pipet.

(**f**) *Preparation of samples for LC analyses*.—Transfer the solutions from either (**d**) or (**e**) to a 10 mL disposable syringe, **B**(**z**), and filter through a 0.45 µm filter, **B**(**w**).

(**g**) *HPLC of sample solutions*.—Use a 100 µL LC glass syringe, **B**(**aa**), to fill the 50 µL injection loop on the LC system, **B**(**u**)(*2*), with filtered sample extracts. For manually deionized samples from **I**(**e**) after filtration step, **I**(**f**), apply solutions via the injection loop, **B**(**u**)(*2*), directly onto the Waters

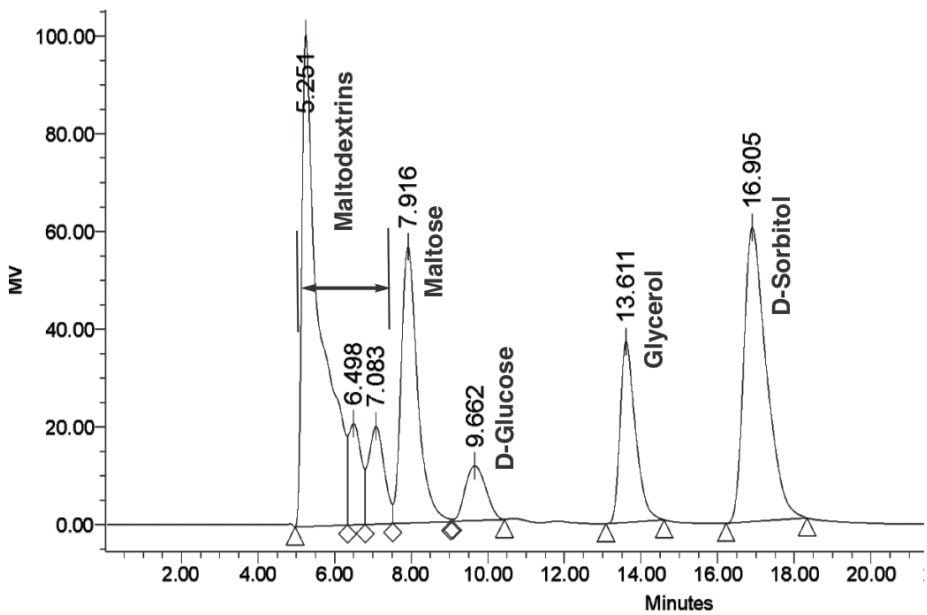

Figure 2011.25E. Chromatography of a mixture of maltodextrins, maltose, D-glucose, D-sorbitol, and glycerol on a Waters Sugar-Pak column. Right arrow shows demarcation between DP 2 (maltose) and DP 3 (higher maltodextrins).

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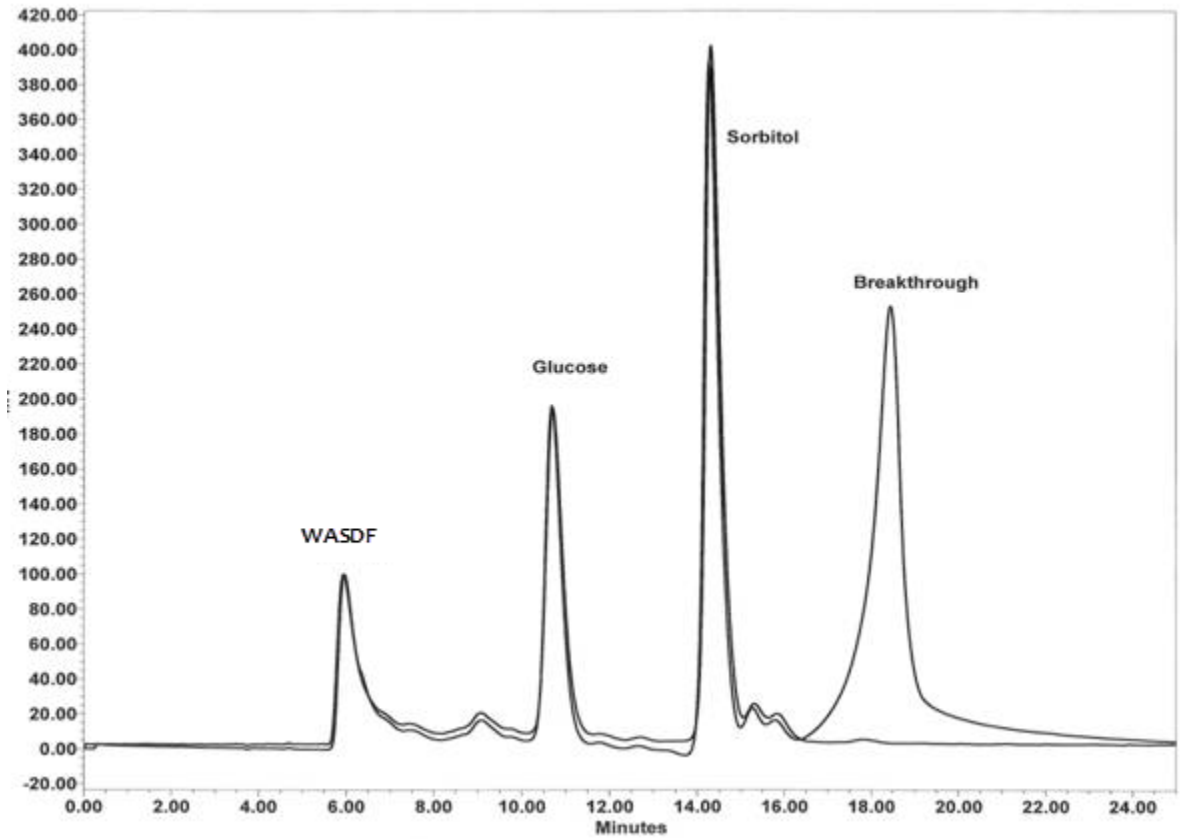

Figure 2011.25F. Chromatogram of typical SDFS in sample on a Sugar-Pak column analyzed before and after the capacity of the deionizing cartridges has been expended. Note the absence of the peak at 18.4 min when the deionizing cartridges are functioning properly, and appearance of the ion peak at 18.4 min indicating the need for replacement of cartridges.

Sugar-Pak column, **B**(**u**)(*3*), using the arrangement shown in Figure **2011.25C**. For samples to be deionized online from **I**(**d**) after filtration step, **I**(**f**), apply solutions to the injection loop and the online desalting guard columns, **B**(**t**), using the arrangement shown in Figure **2011.25D**.

(**h**) *Determine peak area of SDFS (PA-SDFS) and internal standard if used (PA-IS) in chromatograms of sample extracts*.—Record area of all peaks of DP greater than the DP 2/DP 3 demarcation point as PA-SDFS. Record the peak area of internal standard as PA-IS.

J. Calculations for IDF

(**a**) *Blank (B, mg) determination.*

$$B = [(BR1 + BR2)]/2 - P_B - A_B$$

where BR1 and BR2 = blank residue mass (mg) for duplicate IDF blank determinations, respectively, and P_B and A_B = mass (mg) of protein and ash, respectively, determined on first and second blank residues.

(**b**) *IDF*.

$$\text{IDF, mg/100 g} = [(R1 + R2)/2 - P - A - B]/(M1 + M2)/2] \times 100$$

$$\text{IDF, \%} = \text{IDF (mg/100 g)}/1000$$

where R1 = IDF residue mass 1 from M1 in mg; R2 = IDF residue mass 2 from M2 in mg; M1 = test portion mass 1 in g; M2 = test portion mass 2 in g; A = ash mass (mg) from R1; P = protein mass (mg) from R2; B = IDF blank.

K. Calculations for SDFP

(**a**) *Blank (B, mg) determination*.

$$B = [(BR1 + BR2)]/2 - P_B - A_B$$

where BR1 and BR2 = blank residue mass (mg) for duplicate SDFP blank determinations, respectively, and P_B and A_B = mass (mg) of protein and ash, respectively, determined on first and second blank residues.

(**b**) *SDFP*.

$$\text{SDFP, mg/100 g} = [(R1 + R2)/2 - P - A - B]/(M1 + M2)/2] \times 100$$

$$\text{SDFP, \%} = \text{SDFP (mg/100 g)}/1000$$

where R1 = SDFP residue mass 1 from M1 in mg; R2 = SDFP residue mass 2 from M2 in mg; M1 = test portion mass 1 in g; M2 = test portion mass 2 in g; A = ash mass (mg) from R1; P = protein mass (mg) from R2; B = SDFP blank.

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L. Calculations for SDFS

(**a**) *Internal standard method*.—[*Note*: Sorbitol is present in some foods and food products, such as plums (prunes), pears, apples, and polydextrose and may also be added to foods. If sorbitol is present, use of the internal standard will result in artificially low SDFS quantity. In these cases, the external standard method of calculation should be used. Addition of the internal standard is still recommended to use as a retention time reference.]

$$\text{SDFS, mg/100 g} = \text{Rf} \times (\text{Wt-IS, mg}) \times (\text{PA-SDFS})/(\text{PA-IS}) \times 10 \times 100/\text{M}$$

$$\text{SDFS, \%} = \text{SDFS (mg/100 g)}/1000$$

where Rf = response factor calculated in **I**(**c**); Wt-Glu = mass (mg) of D-glucose contained in 1 mL of standard (5, 10, or 20 mg); PA-Glu = peak area D-glucose; Wt-IS is mass of internal standard in 1 mL of sample (i.e., 10 mg per mL; final sample volume, 10 mL); PA-SDFS is the peak area of the water:alcohol soluble dietary fiber; PA-IS is the peak area of the internal standard; 10 is factor to convert from 1 to 10 mL (final sample volume); 100 is factor to convert to 100 g of sample; M is the test portion mass (g) M1 or M2 of the sample whose filtrate was concentrated and analyzed by LC.

(**b**) *External standard method*.

$$\text{SDFS, mg/100 g} = \text{Rf} \times (\text{PA-SDFS}) \times 10 \times 100/\text{M}$$

$$\text{SDFS, \%} = \text{SDFS (mg/100 g)}/1000$$

where Rf = (Wt-Glu)/(PA-Glu); Wt-Glu = mass (mg) of D-glucose contained in 1 mL of standard (5, 10, or 20 mg); PA-Glu = peak area for D-glucose (5, 10, or 20 mg); PA-SDFS is the peak area of the water:alcohol soluble dietary fiber; 10 is factor to convert from 1 to 10 mL (final sample volume); 100 is factor to convert to 100 g of sample; M is the test portion mass (g) M1 or M2 of the sample whose filtrate was concentrated and analyzed by LC.

M. Calculations for SDF

$$\text{SDF, mg/100 g} = \text{SDFP (mg/100 g)} + \text{SDFS (mg/100 g)}$$

$$\text{SDF, \%} = \text{SDF (mg/100 g)}/1000$$

N. Calculation for TDF

$$\text{TDF, \%} = \text{IDF (\%)} + \text{SDF (\%)}$$

O. Determination of TDF Without Fractionation of IDF and SDF

(**a**) *Precipitation of water soluble SDFP in the presence of IDF*.—To each sample enzyme digest, from step **F**(**b**)(*9*), add water to bring the total volume to exactly 71 mL. Then add 279 mL (measured at room temperature) of 95% (v/v) EtOH or IMS, **C**(**a**), preheated to 60°C and mix thoroughly. Allow the SDFP precipitate to form in the beaker with the IDF at room temperature for 60 min.

(**b**) *Filtration setup*.—Tare crucible containing Celite to nearest 0.1 mg. Wet and redistribute the bed of Celite in the crucible, using 15 mL of 78% (v/v) EtOH or IMS, **C**(**b**), from wash bottle. Apply suction to crucible to draw Celite onto fritted glass as an even mat. Discard this filtrate.

(**c**) *Filtration*.—Using vacuum, filter precipitated SDFP:IDF, (**a**), from supernatant through crucible. Using a wash bottle with 78% (v/v) EtOH or IMS, **C**(**b**), quantitatively transfer all

Table 4. Comparable AOAC method data

Method No.	Title	S_r	RSD_r	S_R	RSD_R	HorRat
985.29	Total Dietary Fiber in Foods	0.15–0.99	0.56–66.25	0.27–1.36	1.58–66.25	0.76–17.46
991.42	Insoluble Dietary Fiber in Food and Food Products	0.41–2.82	0.86–0.38	0.62–9.49	3.68–19.44	1.73–8.68
991.43[a]	Total, Soluble, and Insoluble Dietary Fiber in Food and Food Products	0.36–1.06	1.50–6.62	0.85–2.06	1.58–12.17	0.74–4.66
992.16	Total Dietary Fiber	0.18–1.01	1.48–14.73	0.22–2.06	4.13–17.94	1.84–4.62
993.19	Soluble Dietary Fiber in Food and Food Products	0.49–1.15	1.74–5.93	0.79–2.05	2.41–7.01	1.13–2.83
994.13	Total Dietary Fiber (Determined as Neutral Sugar Residues, Uronic Acid Residues, and Klason Lignin)	0.32–2.88	1.80–6.96	0.52–4.90	4.80–11.30	2.32–4.20
2001.03	Dietary Fiber Containing Supplemented Resistant Maltodextrin (RMD)	0.02–1.63	1.33–6.10	0.04–2.37	1.79–9.39	0.77–3.32
2002.02	Resistant Starch in Starch and Plant Materials	0.08–2.66	1.97–4.12	0.21–3.87	4.58–10.9	1.44–3.74
2009.01	Total Dietary Fiber in Foods	0.41–1.43	1.65–12.34	1.18–5.44	4.70–17.97	1.91–6.49

[a] Samples that were not dried and/or desugared only.

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Table 5. Collaborative study data for total dietary fiber (%TDF as measured directly reported by laboratories)

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1	28.51	29.58	9.78	9.51	25.80	25.30	11.51	11.52	20.50	19.85	14.81	14.11	20.90	23.16	17.69	17.70
2	27.42	27.86	10.09	9.53	23.49	23.06	11.75	11.71	19.10	19.02	15.25	15.71	19.65	20.23	17.13	17.46
3	27.56	27.32	9.74	10.10	23.69	24.84	8.49	8.89	20.98	22.15	14.62	15.69	22.48	22.68	17.50	17.50
4	28.07	27.76	11.89	12.88	27.88	29.34	13.13	14.01	23.10	21.77	29.09	27.29	23.25	20.89	17.10	13.19
5	27.33	27.81	9.93	9.97	23.54	24.52	10.04	10.01	20.90	23.50	14.27	15.64	20.25	19.17	15.56	17.14
6	27.79	29.04	9.02	8.55	23.24	23.30	10.12	10.40	19.75	19.97	14.68	14.17	20.74	19.90	11.86	11.73
7	26.67	27.42	8.56	8.45	21.84	22.13	6.20	7.03	18.81	18.51	12.77	12.53	19.23	19.33	15.06	15.12
8	25.70	29.35	10.24	11.25	25.64	24.32	11.68	11.19	22.03	20.32	23.78c[a]	14.63c	20.19	20.19	16.80	16.29
9	28.40	31.80	11.42c	14.71c	26.15	24.72	14.72	15.63	27.26	24.30	35.69c	30.85c	23.56	27.12	19.61	22.06

[a] c = Cochran test outlier on total dietary fiber.

remaining particles to crucible. Retain filtrate and washings, (**c**) and (**d**), for determination of low-molecular-weight SDF, **I**(**d**).

(**d**) *Wash and retain filtrate*.—Using a vacuum, wash residue successively with two 15 mL portions of the following: 78% (v/v) EtOH or IMS, followed by two 15 mL portions of 95% (v/v) EtOH or IMS, followed by two 15 mL portions of acetone. Combine with filtrate from step immediately preceding.

(**e**) Dry crucibles containing residue overnight in 105°C oven.

(**f**) Cool and weigh crucible in desiccator for approximately 1 h. Weigh crucible containing dietary fiber residue and Celite to nearest 0.1 mg. To obtain residue mass, subtract tare weight, i.e., weight of dried crucible and Celite.

(**g**) *Protein and ash determination*.—Residue from one crucible of the duplicates is analyzed for protein, and the second residue of the duplicates is analyzed for ash. Perform protein analysis on residue using Kjeldahl or combustion methods. (Caution should be exercised when using a combustion analyzer for protein in the residue. Celite volatilized from the sample can clog the transfer lines of the unit.) Use 6.25 factor for all cases to calculate mg of protein. For ash analysis, incinerate the second residue for 5 h at 525°C. Cool in desiccator and weigh to nearest 0.1 mg. Subtract crucible and Celite weight to determine ash.

(**h**) Proceed to **I**(**d**) for determination of SDFS.

P. Calculations for SDFP:IDF

(**a**) *Blank (B, mg) determination*.

$$B = [(\mathrm{BR1} + \mathrm{BR2})]/2 - P_B - A_B$$

where BR1 and BR2 = blank residue mass (mg) for duplicate SDFP:IDF blank determinations, respectively, and P_B and A_B = mass (mg) of protein and ash, respectively, determined on first and second blank residues.

(**b**) *SDFP:IDF*.

Table 6. Collaborative study data for total dietary fiber (%TDF as measured directly versus sum of IDF and SDF reported by laboratories that ran both methods)

	Sample															
	A		B		C		D		E		F		G		H	
1 Direct	28.51	29.58	9.78	9.51	25.80	25.30	11.51	11.52	20.50	19.85	14.81	14.11	20.90	23.16	17.69	17.70
1 Sum	28.90	27.82	9.87	9.80	27.05	28.32	11.78	12.02	21.03	20.22	17.12	17.35	22.90	23.04	17.25	17.81
2 Direct	27.42	27.86	10.09	9.53	23.49	23.06	11.75	11.71	19.10	19.02	15.25	15.71	19.65	20.23	17.13	17.46
2 Sum	27.93	29.32	10.61	10.04	24.01	23.56	12.26	12.21	19.62	19.53	15.76	16.85	20.17	20.74	17.64	17.97
3 Direct	27.56	27.32	9.74	10.10	23.69	24.84	8.49	8.89	20.98	22.15	14.62	15.69	22.48	22.68	17.50	17.50
3 Sum	28.87	28.75	9.39	9.95	25.97	29.51	10.06	10.23	24.15	24.94	19.63	21.29	34.48	21.49	22.79	20.47
4 Direct	28.07	27.76	11.89	12.88	27.88	29.34	13.13	14.01	23.10	21.77	29.09	27.29	23.25	20.89	17.10	13.19
4 Sum	27.07	26.45	9.67	10.10	21.97	21.00	12.02	13.31	19.24	18.31	13.95	13.41	20.52	17.38	16.22	16.43
5 Direct	27.79	29.04	9.02	8.55	23.24	23.30	10.12	10.40	19.75	19.97	14.68	14.17	20.74	19.90	11.86	11.73
5 Sum	28.12	29.59	9.24	9.02	23.34	23.54	10.25	10.01	20.63	20.67	*[a]	*	20.40	20.30	12.15	11.53

[a] * = Laboratory reported no result for this sample.

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Table 7. ANOVA results for sum of IDF plus SDF versus TDF measured directly

Source	Sum of squares	Df	Mean square	*F*-ratio[a]	*P*-value
			Main effects		
A:Lab	129.604	4	32.401	10.77	0.0000
B:Method	0.247724	1	0.247724	0.08	0.7747
C:Sample	5629.01	7	804.144	267.26	0.0000

[a] All *F*-ratios are based on the residual mean square error.

$$\text{SDFP:IDF, mg/100 g} = [(R1 + R2)/2 - P - A - B]/(M1 + M2)/2] \times 100$$

$$\text{SDFP:IDF, \%} = \text{SDFP:IDF (mg/100 g)}/1000$$

where R1 = SDFP:IDF residue mass 1 from M1 in mg; R2 = SDFP:IDF residue mass 2 from M2 in mg; M1 = test portion mass 1 in g; M2 = test portion mass 2 in g; A = ash mass (mg) from R1; P = protein mass (mg) from R2; B = blank.

Q. Calculation for TDF

$$\text{TDF, \%} = \text{SDFP:IDF (\%) (from } \mathbf{P}\text{)} + \text{SDFS (\%) (from } \mathbf{L}\text{)}$$

Reference: *J. AOAC Int.* **95**, 824(2012)

Results and Discussion

To simulate food digestion in the small intestine, a combination of gentle shaking combined with enzymatic digestion at 37°C is used. Adjusting the pH of the digestion solution followed by temporary heating to 100°C destroys the amylase and AMG activity and promotes partial denaturation of protein, providing efficient protein digestion after cooling to 60°C. Incorporating the water-insoluble, and the SDF, SDFP, segregation steps, and the liquid chromatographic quantitation of the SDFS as validated in previously adopted *Official Methods of Analysis* completes the assay.

The raw data results of the dietary fiber collaborative study are shown in Tables 1–3 for IDF, SDF, and TDF, respectively.

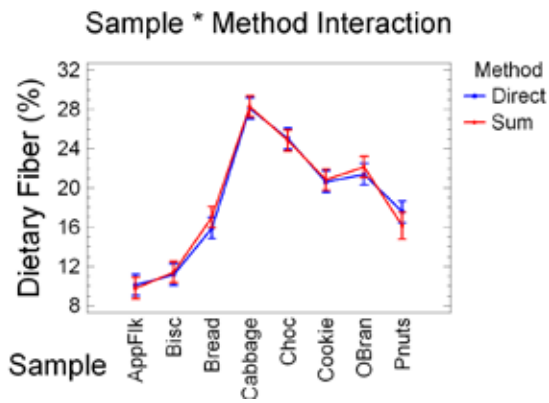

Figure 1. Comparison of TDF by sum of IDF and SDF versus TDF obtained directly.

Cochran and Grubbs outliers are noted directly in the tables. Tables **2011.25A–C** show the statistical results obtained after removal of outliers for IDF, SDF, and TDF, respectively, as described in the statistics section above. Samples for this collaborative study were chosen to be challenging, i.e., with emphasis on quantitating products with resistant starch (RS) and products with nondigestible oligosaccharides. The within-laboratory variability, repeatability SD (s_r), for IDF ranged from 0.13 to 0.71, and the between-laboratory variability, reproducibility SD (s_R), for IDF ranged from 0.42 to 2.24. The within-laboratory variability, s_r, for SDF ranged from 0.28 to 1.03, and the between-laboratory variability, s_R, for SDF ranged from 0.85 to 1.66. The within-laboratory variability, s_r, for TDF ranged from 0.47 to 1.41, and the between-laboratory variability, s_R, for TDF ranged from 0.95 to 3.14. Compared to the dietary fiber statistical results for previously adopted *Official Methods* for dietary fiber shown in Table 4, this level and range of variability are similar to those of other dietary fiber methods, influenced by the significant number of technique-dependent manual operations, each of which contributes to the overall variability of the final results. Repeatability, reproducibility, and HorRat values were within the range of performance characteristics typically found for dietary fiber methods, in which a significant number of manual steps are necessary to perform the assay. In previously adopted methods, the between-laboratory variability, s_R, ranged from 0.04 to 9.49

Table 8. Collaborative study data for SDF (%SDF as measured using online deionization)

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1	2.68	2.43	4.32	4.78	17.30	18.10	10.49	10.43	12.50	12.40	2.34	2.53	11.19	11.54	8.32	8.73
2	4.43	4.18	4.86	4.85	16.48	15.69	5.95	6.11	12.62	12.55	3.12	3.23	10.96	11.03	6.76	6.53
3	2.88	3.79	5.33	4.93	15.94	14.86	9.61	9.25	11.61	11.24	3.25	3.33	11.22	10.56	5.39	6.07
4	3.60	3.88	4.57	3.86	14.54	15.15	9.72	9.49	10.63	10.99	2.36	2.82	10.41c[a]	12.22c	7.26	7.55
5	2.94	3.32	5.04	4.88	16.59	15.49	6.95	5.44	11.17	11.69	2.63	2.51	10.47	10.39	7.93	8.05
6	3.40	3.90	5.34	5.66	17.61	17.69	9.54	10.19	12.67	13.03	3.57	3.67	11.22	11.26	6.27	6.41

[a] c = Cochran test outlier on soluble dietary fiber.

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Table 9. Collaborative study data for TDF (%TDF as measured using online deionization)[a]

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1	28.58	27.53	9.62	9.78	27.40	28.90	11.69	11.63	21.00	20.60	16.24	16.73	22.99	23.24	18.62	19.03
2	29.82	29.36	9.95	9.76	28.88	28.84	7.11dg	7.30dg	20.93	20.86	13.71	14.04	24.28	24.49	17.00	16.79
3	30.16	31.08	10.76	10.69	26.32	26.02	11.30	11.15	20.81	20.57	15.85	16.34	24.47	24.53	17.03	17.16
4	29.84	30.84	9.66	8.76	25.23	25.93	10.90	11.03	19.21	19.96	13.88	13.69	21.81c	24.07c	17.32	17.95
5	27.53	29.32	9.95	9.21	24.23	22.49	8.21dg	6.77dg	18.98	19.57	14.69	15.83	19.95	20.64	17.33	17.39
6	30.72	30.99	10.50	10.81	30.02	29.30	10.63	11.37	21.41	21.76	14.60	15.17	23.01	22.79	16.88	16.68

[a] dg = Double Grubbs test outlier on insoluble dietary fiber; c = Cochran test outlier on total dietary fiber.

and the between-laboratory relative variability, RSD_R, ranged from 1.58 to 66.25%. This is due to the fact that all dietary fiber methods to date comprise a significant number of technique-dependent manual operations, each of which contributes to the overall variability of the final results. Table 4 shows the statistical characteristics of the various AOAC methods. The statistical characteristics of this method, which combines steps from AOAC Methods **985.29** (and its extensions, **991.42** and **993.19**), **991.43**, **2001.03**, and **2002.02**, lie within the ranges of the statistical characteristics of the current *Official Methods* for dietary fiber.

A number of laboratories participating in the study determined TDF directly without quantitating the insoluble and soluble dietary fiber fractions. The data provided by those laboratories are shown in Table 5. The statistical results for measuring TDF directly are shown in Table **2011.25D**. Within-laboratory replication (s_r; 0.39–1.26) and between-laboratory replication (s_R; 1.24–5.27) are similar to the data obtained by summing IDF and SDF and are within the ranges for previous dietary fiber methods shown in Table 4. Five laboratories carried out the analysis by both approaches, i.e., TDF by summing IDF and SDF as well as measuring TDF directly. A comparison of the results is shown in Table 6. Analysis of variance (ANOVA; Type II-Sum of Squares) was performed on the comparative data (Table 7). The ANOVA showed no statistical difference between the two method approaches. The results shown in Figure 1 are in very good agreement. The statistical data show a very good comparison to the data from the collaborative study for AOAC **2009.01** (method is similar in all respects, except that the current method uses updated equipment for higher productivity), indicating that the method described here provides results with the same level of confidence as AOAC **2009.01**. The analyst thus has a choice between this method and AOAC **2009.01** for measuring TDF directly.

Six of the laboratories involved in the study were able to equip themselves with the online deionization option to remove salts from the sample digests and filtrates as part of the HPLC assay of SDFS versus manual deionization prior to HPLC assay. Results for the SDF and TDF obtained using direct online deionization, along with statistical data, are shown in Tables 8, **2011.25E**, 9, and **2011.25F**. The within-laboratory variability, s_r, for SDF using online deionization ranged from 0.16 to 0.58, and the between-laboratory variability, s_R, for SDF using online deionization ranged from 0.40 to 1.99. The within-laboratory variability, s_r, for TDF using online deionization ranged from

Table 10. Comparison of SDF online versus manual deionization

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1 Online	2.68	2.43	4.32	4.78	17.30	18.10	10.49	10.43	12.50	12.40	2.34	2.53	11.19	11.54	8.32	8.73
1 Manual	3.00	2.72	4.57	4.80	16.95	17.52	10.58	10.82	12.53	12.02	3.22	3.15	11.10	11.34	6.95	7.51
2 Online	4.43	4.18	4.86	4.85	16.48	15.69	5.95	6.11	12.62	12.55	3.12	3.23	10.96	11.03	6.76	6.53
2 Manual	4.98	5.86	6.75	7.27	17.84	18.30	11.17	11.01	13.61	14.25	4.02	6.88	12.14	12.76	10.24	9.24
3 Online	2.88	3.79	5.33	4.93	15.94	14.86	9.61	9.25	11.61	11.24	3.25	3.33	11.22	10.56	5.39	6.07
3 Manual	3.05	4.31	5.76	5.80	17.17	17.08	10.12	11.66	12.66	12.67	10.26	4.58	11.91	11.57	8.13	7.82
4 Online	3.60	3.88	4.57	3.86	14.54	15.15	9.72	9.49	10.63	10.99	2.36	2.82	10.41	12.22	7.26	7.55
4 Manual	3.71	3.69	5.11	4.69	16.19	16.12	10.51	9.99	11.67	11.69	2.31	3.17	10.16	11.14	7.59	6.52
5 Online	2.94	3.32	5.04	4.88	16.59	15.49	6.95	5.44	11.17	11.69	2.63	2.51	10.47	10.39	7.93	8.05
5 Manual	3.34	3.32	5.70	5.71	16.37	16.56	11.00	10.88	11.81	11.65	3.70	3.53	10.69	10.49	8.24	8.63
6 Online	3.40	3.90	5.34	5.66	17.61	17.69	9.54	10.19	12.67	13.03	3.57	3.67	11.22	11.26	6.27	6.41
6 Manual	3.98	4.33	6.16	6.58	17.89	17.95	10.80	11.52	13.53	13.61	4.24	4.43	11.63	12.69	8.68	8.17

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Table 11. Comparison of TDF online versus manual deionization

	Sample															
Lab No.	A		B		C		D		E		F		G		H	
1 Online	28.58	27.53	9.62	9.78	27.40	28.90	11.69	11.63	21.00	20.60	16.24	16.73	22.99	23.24	18.62	19.03
1 Manual	28.90	27.82	9.87	9.80	27.05	28.32	11.78	12.02	21.03	20.22	17.12	17.35	22.90	23.04	17.25	17.81
2 Online	29.82	29.36	9.95	9.76	28.88	28.84	7.11	7.30	20.93	20.86	13.71	14.04	24.28	24.49	17.00	16.79
2 Manual	30.37	31.04	11.85	12.61	30.24	31.45	12.33	12.20	21.93	22.57	14.60	17.69	25.46	26.81	20.48	19.50
3 Online	30.16	31.08	10.76	10.69	26.32	26.02	11.30	11.15	20.81	20.57	15.85	16.34	24.47	24.53	17.03	17.16
3 Manual	30.33	31.59	11.19	11.56	27.54	28.24	11.80	13.56	21.86	22.00	22.86	17.61	25.16	25.56	19.78	18.91
4 Online	29.84	30.84	9.66	8.76	25.23	25.93	10.90	11.03	19.21	19.96	13.88	13.69	21.81	24.07	17.32	17.95
4 Manual	29.95	30.66	10.19	9.60	26.88	26.91	11.69	11.53	20.25	20.65	13.84	14.04	21.56	22.99	17.66	16.92
5 Online	27.53	29.32	9.95	9.21	24.23	22.49	8.21	6.77	18.98	19.57	14.69	15.83	19.95	20.64	17.33	17.39
5 Manual	27.93	29.32	10.61	10.04	24.01	23.56	12.26	12.21	19.62	19.53	15.76	16.85	20.17	20.74	17.64	17.97
6 Online	30.72	30.99	10.50	10.81	30.02	29.30	10.63	11.37	21.41	21.76	14.60	15.17	23.01	22.79	16.88	16.68
6 Manual	31.30	31.42	11.32	11.73	30.30	29.56	11.88	12.70	22.26	22.34	15.26	15.93	23.42	24.22	19.29	18.45

0.24 to 0.73, and the between-laboratory variability, s_R, for TDF using online deionization ranged from 0.37 to 2.41. Thus, online deionization showed slightly better precision than manual deionization. Tables 10 and 11 present a comparison of the SDF and TDF values obtained by manual versus online deionization. These results are plotted in Figures 2 and 3 for SDF and TDF, respectively. As seen in Figures 2 and 3, the online deionization showed a small (about 5% relative) bias to a lower value for some of the samples for both SDF and TDF. Further research is under way to understand the reason for this difference.

Twelve of the laboratories provided data for calculating the SDF and the TDF by the external standard method. Results using the external standard method of calculation for SDF and TDF, along with statistical summaries of these results, are shown in Tables 12, **2011.25G**, 13, and **2011.25H**. As can be seen, the results of the external and internal standard methods agree well, with the external standard method having less variability in some cases and the internal standard method having less variability in others.

Collaborators' Comments

One collaborating laboratory commented that new deionization columns should be used each time because HPLC performance was noisier when the deionization columns were regenerated and reused. While the regeneration of the deionizing columns described in the method appears to be effective in most laboratories, we believe that reuse of resins is not advisable on an ongoing basis. With the new recommended column to contain the resins and the reduced quantity of resin per sample, column repacking is simple and the cost of resin is insignificant.

One collaborator reported difficulty obtaining an HPLC pump capable of precise flow (exhibited noisy baselines) at an EDTA solution flow rate of 0.05 mL/min; therefore, the work was done with a flow rate of 0.1 mL/min. The results were in line with those of the other laboratories in the study. Since the analytical column should be calcium-saturated, the excess calcium salt flow should not be an issue.

Another collaborator's comment concerned the use of Duran bottles rather than beakers for the enzymatic digestion steps of the method. The cap on the Duran bottles retains liquid that must be carefully recovered during filtering. It is also more difficult to transfer traces of dietary fiber from the neck of the bottle to the filter than it is with a beaker, as used in AOAC Methods **985.29** and **991.43**. Duran bottles were chosen, in conjunction with the shaker, to ensure that the entire sample being tested properly contacts the enzymes during digestion. Use of the bottle and shaker does not allow rings of sample to form above the solution and keeps the mix homogeneous during the entire 16 h digestion. An ideal solution would be a vessel that was as easy to use as the beakers used in Method **985.29** for manual operations while ensuring complete enzyme interaction with the sample during digestion. The Study Directors agreed there is a limitation in using the Duran bottles, so a switch was made to Fisherbrand bottles, **B**(**b**), which are now part of the recommended procedure.

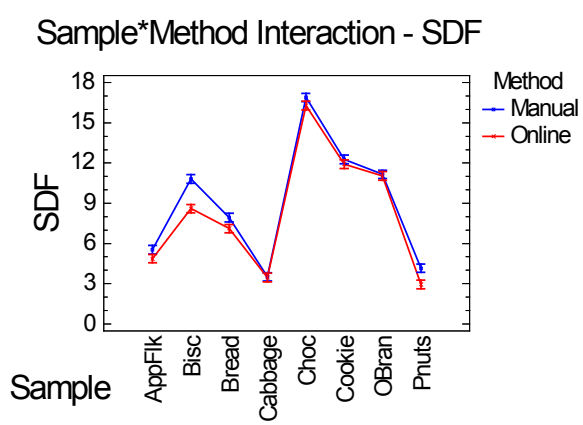

Figure 2. Comparison of SDF obtained by manual deionization procedure versus online deionization procedure.

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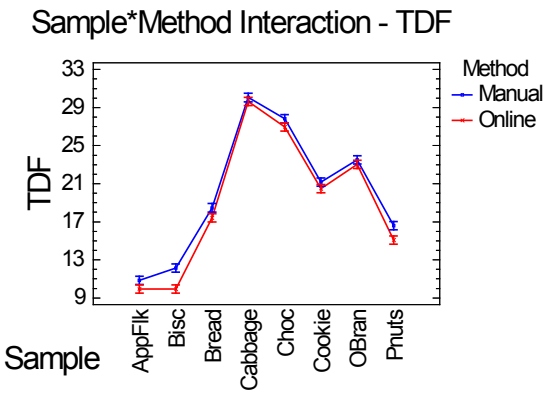

Figure 3. Comparison of TDF obtained by manual deionization versus online deionization.

One collaborator commented that SDFS results might be biased due to the use of sorbitol in the internal standard method. The collaborator stated that sorbitol is naturally present in some foods and food products, primarily fruits; the collaborator also encountered difficulty in properly integrating the peak area of the sorbitol, which is not always baseline-resolved from neighboring peaks. The Study Directors are aware of the presence of sorbitol in a number of fruits, such as dried plums, prunes, apples (fresh and dried), and pears (fresh and dried). Sorbitol is also present at varying levels in polydextrose, a synthetic carbohydrate polymer that resists digestion and is used as a food bulking agent to provide reduced calories. The Study Directors agree that for these foods and for foods with sorbitol added, low analytical results for SDF will be obtained. Therefore, a note has been added to the method to this effect. The Study Directors did not encounter problems in integrating the peak area of sorbitol. If a more appropriate internal standard can be identified this will be evaluated.

Another collaborator preferred to use the glass column-based deionization procedure of **2009.01**, rather than the disposable plastic column-based deionization procedure provided with the method. The laboratory is comfortable with that procedure, and noted that occasionally the plastic columns clog. The Study Directors do not see an issue with using glass versus plastic for this procedure, and concluded that a single column with mixed bed resin was preferable.

One collaborator commented that the fructosyl trisaccharide (F3) derived from acid or enzymic hydrolysis of inulin elutes from the Sugar-Pak column slightly later than maltose and sucrose, and at the same spot as lactose. At the time of the study, this trisaccharide was not available, and because the amount present in the Raftilose P-95 product available to the Study Directors contained a relatively low amount of this trisaccharide, the problem was not noticed. Further studies by the Study Directors have resolved this problem as part of sample preparation. Basically, an aliquot of the sample containing F3 is incubated with a thermostable α-glucosidase which hydrolyzes both sucrose and maltose, allowing the amount of F3 to be determined accurately. For samples containing lactose, incubation with β-galactosidase removes this disaccharide. The determined area of F3 is then added to the area of oligosaccharides eluting earlier than maltotriose. A mixture of α-glucosidase/ β-galactosidase is available from Megazyme International, along with instructions on use (Cat. No. E-SUCRBG). An alternative solution is chromatography of the oligosaccharide mixture on 2-size exclusion LC columns (TSK-Gel 30 cm × 7.8 mm) connected in series (Sigma Aldrich Part No. 808020), as described in AOAC *Official Method* **2009.01**.

Table 12. Collaborative study SDF data calculated using external standard approach[a]

	Sample															
Lab	A		B		C		D		E		F		G		H	
1	2.92	2.62	4.28	4.49	14.80	16.20	9.79	10.40	11.70	11.60	3.20	2.98	10.95	11.13	6.31	7.03
2	3.35	3.31	5.60	5.68	16.47	16.49	9.04	8.78	10.65	10.18	3.03	2.73	10.25	10.06	8.02	8.45
3	4.54c	8.16c	8.31	7.37	19.61	18.41	13.98	12.88	16.31	16.73	6.57	5.80	12.39c	15.34c	11.78	13.56
4	5.04	6.41	6.28	6.72	15.33	17.57	11.58	11.35	13.29	14.19	4.04	6.46	13.61	12.12	10.19	9.22
5	2.77	3.65	4.98	4.41	11.19	13.23	5.56	7.53	9.13	7.23	2.39	3.69	11.29	11.00	5.65	5.87
6	5.19	5.32	5.36	7.34	15.15	17.55	11.16	12.21	13.38	13.57	4.97	3.92	12.45	12.50	9.75c	13.42c
7	4.48	3.59	5.56	5.99	16.74	18.09	10.39	10.32	12.21	11.85	3.18	3.65	11.16	11.91	8.11	7.89
8	4.31	4.32	4.04	4.27	16.03	16.63	9.82	8.74	13.45	13.82	3.40	3.81	11.00	10.94	8.91	7.73
9	3.73	4.06	5.15	5.83	13.99	14.27	8.55	9.99	11.60	12.65	3.91	4.18	11.46	11.96	7.40	7.01
10	4.58	4.11	5.74	4.97	19.82	15.28	11.39	9.37	14.84	12.69	4.45	4.54	11.98	12.13	7.46	7.48
11	3.74	3.65	5.47	4.81	13.42	16.74	10.99	9.70	12.04	12.06	2.27	3.07	10.18	11.25	7.47	6.24
12	2.37	3.12	4.61	4.64	15.80	16.24	9.09	8.91	12.59	12.68	*[b]	*	10.04	10.03	2.55dg	2.48dg

[a] c = Cochran test outlier on insoluble dietary fiber; sg = Single Grubbs test outlier on insoluble dietary fiber; dg = Double Grubbs test outlier on insoluble dietary fiber.

[b] * = Laboratory reported no result for this sample.

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Table 13. Collaborative study TDF data calculated using external standard approach[a]

	Sample															
Lab	A		B		C		D		E		F		G		H	
1	28.82	27.72	9.58	9.49	24.90	27.00	10.99	11.60	20.20	19.80	17.10	17.18	22.75	22.83	16.61	17.33
2	27.94	29.31	10.51	10.01	24.11	23.49	10.30	10.11	18.46	18.06	15.09	16.05	19.73	20.31	17.42	17.79
3	31.14	33.17	13.03	11.88	26.90	27.44	15.19	13.99	24.11	24.38	17.88	16.25	23.29	26.10	20.64	22.13
4	30.43	31.59	11.37	11.63	27.72	30.73	12.75	12.53	21.60	22.50	14.63	17.26	26.94	25.59	20.43	19.48
5	30.05	30.93	10.40	10.17	21.57	24.38	7.24	9.42	18.33	16.56	14.99	16.71	24.54	24.98	17.30	16.97
6	32.10	35.84	9.48	11.63	25.54	26.29	11.70	12.74	21.07	24.65	42.91sg	42.24sg	28.55	29.43	25.51dg	28.47dg
7	29.18	29.09	10.91	11.15	25.52	26.79	11.62	11.83	20.65	20.50	14.72	15.11	23.09	23.90	18.42	18.71
8	28.26	28.76	8.47	8.52	23.30	24.13	10.53	9.02	21.63	21.48	13.70	14.19	20.95	21.04	17.81	16.50
9	31.04	31.15	10.31	10.98	26.40	25.88	9.63	11.17	20.34	21.38	14.94	15.67	23.24	23.49	18.00	17.28
10	28.21	28.11	10.55	9.71	27.99	23.70	12.40	10.43	22.83	20.87	15.94	16.37	24.21	24.05	16.81	17.25
11	29.98	30.62	10.55	9.71	24.11	27.53	12.17	11.24	20.62	21.03	13.80	13.93	21.58	23.10	17.54	16.64
12	28.05	29.59	9.22	9.02	23.39	23.54	10.26	10.01	20.66	20.68	*[b]	*	20.37	20.30	12.11dg	11.53dg

[a] c = Cochran test outlier on insoluble dietary fiber; sg = Single Grubbs test outlier on insoluble dietary fiber; dg = Double Grubbs test outlier on insoluble dietary fiber.

[b] * = Laboratory reported no result for this sample.

Recommendations

The Study Directors recommend this method be adopted as an Official First Action Method of AOAC INTERNATIONAL.

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References

(1) *Code of Federal Regulations* (2011) Vol. 2, Title 21 Part 101.81, U.S. Government Printing Office, Washington, DC, pp 141–145

(2) DeVries, J.W., & Rader, J.I. (2005) *J. AOAC Int.* **88**, 1349–1366

(3) Trowell, H. (1972) *Atherosclerosis* **16**, 138–140. http://dx.doi.org/10.1016/0021-9150(72)90017-2

(4) Trowell, H. (1972) *Am. J. Clin. Nutr.* **25**, 926–932

(5) Trowell, H.C. (1974) *Lancet* **1**, 503. http://dx.doi.org/10.1016/S0140-6736(74)92802-5

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(6) Trowell, H.C., Southgate, D.A.T., Wolever, T.M.S., Leeds, A.R., Gassull, M.A., & Jenkins, D.J.A. (1976) *Lancet* **1**, 967. http://dx.doi.org/10.1016/S0140-6736(76)92750-1
(7) Prosky, L., Asp, N.G., Furda, I., DeVries, J.W., Schweizer, T.F., & Harland, B.F. (1985) *J. Assoc. Off. Anal. Chem.* **68**, 677–679
(8) *Official Methods of Analysis* (2005) 18th Ed., AOAC INTERNATIONAL, Gaithersburg, MD
(9) *Official Methods of Analysis* (2010) 18th Ed., Rev. 3, AOAC INTERNATIONAL, Gaithersburg, MD
(10) ALINORM 09/32/26, http://www.codexalimentarius.net/web/archives/jsp
(11) McCleary, B.V. (2007) *Anal. Bioanal. Chem.* **389**, 291–308. http://dx.doi.org/10.1007/s00216-007-1389-6
(12) Post, B.E., Marshak, M.R., & DeVries, J.W. (2010) *J. AOAC Int.* **93**, 234–242

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